

Adaptive Conformal Inference through the Lens of Blackwell Approachability

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Abstract

This article considers an online version of conformal inference, called adaptive conformal inference [ACI] and introduced by Gibbs and Candès (2021): prediction sets are issued sequentially, after observing features and before the outcomes are revealed. These sets are evaluated both in terms of validity (the fraction of rounds where the outcome was lying in the prediction set) and efficiency (the average lengths of the prediction sets). The two criteria point to different directions (validity favors larger sets). We also target a wide range of scenarios, with exchangeable data and arbitrary data (lack of any stochastic guarantees) as two extremes. A series of existing strategies for ACI typically guarantee that empirical coverage converges to the desired level for arbitrary sequences, but they generally lack simultaneous efficiency guarantees. To provide a unified study, we first formulate ACI as a repeated two-player game with finite action sets and vector-valued payoffs encoding validity and efficiency. Building on this reformulation, we introduce a strategy based on Blackwell approachability and on its opportunistic extension by Bernstein et al. (2014) that ensures validity while adapting the efficiency of the prediction intervals to the underlying degree of stochasticity of the opponent player. The resulting guarantee is “best of many worlds”: it recovers the relevant efficiency guarantees in exchangeable and adversarial settings, and provides guarantees in intermediate settings that arise in typical applications such as the forecasting of time series.

Keywords: Adaptive conformal inference; efficient prediction intervals; Blackwell opportunistic approachability.

1 Introduction

Conformal inference (Vovk et al., 2005) is a general framework for constructing prediction sets with finite-sample coverage guarantees under the assumption of exchangeability. This assumption, however, is often violated in sequential or time-dependent settings, such as the forecasting of time series, where the distribution of observations may evolve over time.

To address this limitation, Gibbs and Candès (2021) introduced a setting (and an algorithm) called adaptive conformal inference [ACI], which extends conformal inference beyond exchangeability. The underlying idea is to sequentially update the (levels of the) conformal sets in response to observed outcomes to ensure that empirical coverage converges to the desired level for arbitrary sequences. The ACI algorithm and its extensions satisfy this property, called *validity*, but they generally lack theoretical guarantees on the *efficiency*

of the prediction sets, notably for prediction intervals, for which efficiency is measured in terms of their average lengths.

In this work, we address validity and efficiency jointly by formulating adaptive conformal inference as a repeated two-player finite game with vector-valued payoffs that simultaneously encode these two objectives. Building on this perspective, we design a strategy that ensures validity while achieving a setting-dependent notion of efficiency. Our approach relies on an extension of approachability called opportunistic approachability (Bernstein et al., 2014), and is “best of many worlds” in the sense that this notion of efficiency recovers the relevant guarantees in the extreme settings formed by exchangeable data and adversarial sequences, while offering guarantees in intermediate settings that arise in practical applications such as the forecasting of time series.

1.1 Related Works

ACI and online conformal prediction. ACI maintains coverage by updating the conformal sets via online gradient descent (Gibbs and Candès, 2021). Subsequent works largely adopted this online optimization perspective, with a particular emphasis on the step-size schedule. Zaffran et al. (2022) resorts to aggregation of experts, Angelopoulos et al. (2024) study decaying step sizes, and both Gibbs and Candès (2024) and Bhatnagar et al. (2023) establish connections to adaptive regret minimization.

Within this literature, two parameterizations have emerged. The original ACI updates the miscoverage levels α_t used to compute empirical quantiles of conformity scores, whereas online conformal prediction updates the conformity threshold directly, bypassing the quantile computation (Bhatnagar et al., 2023; Angelopoulos et al., 2024). Despite this distinction, both approaches rely on the same underlying online optimization principle, suggesting that the asymptotic validity achieved depends only weakly on the specific construction of conformal sets (Szabadváry and Löfström, 2025). Other extensions beyond exchangeability include proportional-integral-derivative [PID] control, which combines quantile tracking, error integration, and quantile forecasting to adapt the prediction sets online (Angelopoulos et al., 2023), and reweighting schemes that assign larger weights to recent observations in the empirical distribution of conformity scores (Barber et al., 2023). Closer to our approach, Cauchois et al. (2024) handle distribution shifts by estimating their magnitude and by adjusting the targeted miscoverage levels accordingly.

Validity and something else. Recent work has aimed to extend the guarantees offered by ACI, typically by offering strategies that guarantee both validity and something else. For instance, Angelopoulos et al. (2025) introduce the notion of gradient equilibrium for which ACI is a special case, and show that the broader class of strategies at gradient equilibrium enjoy coverage properties. Ramalingam et al. (2025) investigate the relationship between pinball-loss regret and validity and provide strategies controlling both notions. Pinball-loss regret is frequently used as a proxy for efficiency, but lacks interpretability in terms of the average length of prediction intervals.

Other works focus more directly on efficiency, measured by the average lengths of the prediction intervals output. Zaffran et al. (2022) analyze the efficiency of the original ACI procedure in the exchangeable and autoregressive settings, and show that it is generally suboptimal in terms of efficiency (see Section 5.1 for a detailed discussion on their results).

In the adversarial setting, Srinivas (2026) establish impossibility results, proving that no nontrivial efficiency guarantees are attainable without additional assumptions. This result motivates the consideration of the assumption of Q -restricted opponent introduced by Bernstein et al. (2014), for which we build a meaningful theory of efficiency under validity constraint.

The key tool to do so is related to repeated finite games and is called approachability theory. We review it next.

Approachability theory. Blackwell (1956) introduced the concept of approachability as the possibility, for a learner, to force the average payoffs to converge to a given target set $\mathcal{C}$, irrespective of the opponent. Approachability is a foundational tool in the analysis of repeated finite games and characterizes tractable online learning problems. It was used to show the existence of no-regret strategies (Hart and Mas-Colell, 2000, see also Cesa-Bianchi and Lugosi, 2003 and Abernethy et al., 2011) and of calibrated forecasters (see Foster, 1999, and see also Mannor and Stoltz, 2010).

In the context of adaptive conformal inference, Marx et al. (2025) mention that asymptotic validity can be seen as an approachability problem, with no further analysis of efficiency.

Approachability is generally studied through the lens of the primal condition, which provides an explicit approachability strategy. Our focus, however, lies on the dual condition instead, and is a response-based approachability strategy. Response-based strategies have been studied by Bernstein and Shimkin (2015) (see also Mannor et al., 2014), and we rely especially on the opportunistic approachability theory developed by Bernstein et al. (2014). Opportunistic approachability is a calibration-based approachability algorithm (see Perchet, 2009 for a general study of such algorithms) allowing the learner to take advantage of potential restrictions in the adversary’s play.

1.2 Contributions and Outline

Our contributions are twofold.

(i) A game-theoretical formulation of adaptive conformal inference [ACI]. We first formulate ACI as a repeated game with continuous action sets and then provide a reformulation thereof as a repeated game with finite action sets and vector-valued payoffs that encode simultaneously the validity and efficiency criteria.

(ii) Efficiency guarantees under validity constraints. We then state a strategy, which is an instance of the strategy by Bernstein et al. (2014), that achieves the validity guarantee for any data (be it stochastic or picked by a malicious opponent) while providing a “best of many worlds” efficiency guarantee. Indeed, while ignoring the degree of stochasticity of the online data, this strategy recovers the relevant guarantees in exchangeable and adversarial settings, as well as in intermediate regimes that arise in typical applications such as the forecasting of time series.

The rest of the paper is organized as follows. Section 2, introduces the setting and reformulates ACI as a repeated game with finite action sets and vector-valued payoffs. Section 3, presents the validity and efficiency objectives and provides informal statements of the main theoretical results. Section 4 is devoted to presenting and analyzing in general

the opportunistic approachability strategy that achieves these objectives. Section 5 focuses on more specific examples of settings encompassed by the framework introduced, illustrating its practical relevance. Section 6 mentions a computationally more tractable variant of the strategy based on ε -calibrated forecasters, and establishes the corresponding theoretical guarantees. Finally, the appendices provide the proofs of all results stated in the main body, as well as some background on (opportunistic) approachability and calibration.

Notation. For two integers $1 \leq n \leq m$, let $\llbracket n, m \rrbracket = \{n, n+1, \dots, m\}$. For an integer $n \geq 1$, let $\llbracket n \rrbracket = \llbracket 1, n \rrbracket = \{1, \dots, n\}$. We denote by $\lceil x \rceil$ the upper integer part of the real number x . The Lebesgue measure on $\mathbb{R}$ is denoted by $\mathfrak{L}$. The Dirac mass on a point z is denoted by δ_z . The indicator function of an event A is denoted by $\mathbf{1}_A$. The distance over $\mathbb{R}^d$ is the Euclidean norm $\|\cdot\|$, and the corresponding inner product is denoted by $\langle \cdot, \cdot \rangle$. The convex hull generated by a set S is denoted by $\text{conv}(S)$. The interior of a set S is denoted by $\text{int}(S)$ and its closure by $\text{cl}(S)$. For $u \in \mathbb{R}^d$ and a closed subset $S \subseteq \mathbb{R}^d$, the distance of u to S is defined by $d(u, S) = \min_{s \in S} \|u - s\|$. A sequence $(u_t)_{t \geq 1}$ converges to S , which we denote $u_t \rightarrow S$, if $d(u_t, S) \rightarrow 0$.

The set of all probability measures on a set S is denoted by $\Delta(S)$. The total variation distance between two probability distributions $\mathcal{S}$ and $\mathcal{V}$ is denoted by $d_{\text{TV}}(\mathcal{S}, \mathcal{V})$. When the underlying distribution is ambiguous, we write $\mathbb{P}_{s \sim \mathcal{S}}$ and $\mathbb{E}_{s \sim \mathcal{S}}$ to denote the probability and the expectation with respect to an independent draw s from $\mathcal{S}$. We identify the set of signed distributions over sets of d elements with the simplex of $\mathbb{R}^d$.

2 First Contribution:

Adaptive Conformal Inference [ACI] as a Repeated Game

We (ultimately) work in a sequential prediction setting. At each round $t \in [T]$, after observing vector-valued covariates $x_t \in \mathcal{X}$, where $\mathcal{X} \subseteq \mathbb{R}^p$, the learner outputs a prediction set $C(x_t)$ for the unobserved scalar outcome $y_t \in \mathbb{R}$, which is then revealed before proceeding to the next round. A historical dataset $(x_t, y_t)_{t \in \mathcal{D}}$, indexed by a finite set $\mathcal{D}$, is available and used to perform conformal inference and construct the prediction sets. Note the different sets of indices, $\mathcal{D}$ and $[T]$, for historical and test data.

We first recall two settings of conformal inference: a classic batch setting known as split conformal prediction, where prediction sets are actually constructed simultaneously for the batch $[T]$ of test points, and an online setting known as adaptive conformal inference, where prediction sets are produced sequentially as new outcomes are revealed. Although the batch setting is not the focus of this paper, we briefly recall it for the sake of a more pedagogical exposition.

2.1 Batch Setting of Conformal Inference: Split Conformal Prediction [SCP]

In split conformal prediction (Vovk et al., 2005; Lei et al., 2018), the historical data is split into a training set, indexed by $\mathcal{D}_{\text{train}}$ and a calibration set, indexed by $\mathcal{D}_{\text{calib}}$, of respective cardinalities T_{train} and T_{calib} . A regression function $\hat{\mu}$ is learned on the training set while non-conformity scores $\{s_t : t \in \mathcal{D}_{\text{calib}}\}$, measuring how well y_t is predicted by $\hat{\mu}(x_t)$, are computed on the calibration set. We focus on the standard choice of absolute residuals as non-conformity scores, which leads to conformal sets that are intervals. Thus, for all

$t \in \mathcal{D}_{\text{calib}}$, the non-conformity score is $s_t = |y_t - \hat{\mu}(x_t)|$ and, assuming that the scores are almost surely distinct, we denote their order statistics by $s_{(1)} < \dots < s_{(T_{\text{calib}})}$. We further set $s_{(0)} = 0$ and $s_{(T_{\text{calib}}+1)} = L$, where $L < \infty$ is a known upper bound on the absolute residuals such that $s_{(T_{\text{calib}})} < L$; see Remark 2.

We denote the batch of test points by $(x_t, y_t)_{t \in \mathcal{D}_{\text{test}}}$, where $\mathcal{D}_{\text{test}} = [T]$. For each test point, we define the corresponding non-conformity score as $s_t = |y_t - \hat{\mu}(x_t)|$, following the same definition as for the calibration scores.

For each $t \in [T]$ and for a given miscoverage rate $\alpha \in [0, 1]$, the conformal prediction interval is then

$$C_{1-\alpha}(x_t) = \left[\hat{\mu}(x_t) \pm s_{(\lceil (T_{\text{calib}}+1)(1-\alpha) \rceil)} \right].$$

By construction, conformal prediction intervals are nested in α , i.e., $1 - \alpha \leq 1 - \alpha'$ implies $C_{1-\alpha}(x_t) \subseteq C_{1-\alpha'}(x_t)$, a direct consequence of $s_{(\lceil (T_{\text{calib}}+1)(1-\alpha) \rceil)} \leq s_{(\lceil (T_{\text{calib}}+1)(1-\alpha') \rceil)}$.

Fact 1 *For all $x \in \mathcal{X}$, the family $(C_{1-\alpha}(x))_{\alpha \in [0,1]}$ of conformal intervals has $C_0(x) = \emptyset$ as smallest element, with length $\mathfrak{L}(C_0(x)) = 0$, and $C_1(x) = [\hat{\mu}(x) \pm L]$ as largest element, with length $\mathfrak{L}(C_1(x)) = 2L$. These two extreme cases corresponding to $1 - \alpha \in \{0, 1\}$ will play a role in the subsequent sections.*

We can now state the central result from Vovk et al. (2005), that holds under a stochastic assumption: if calibration scores $(s_t)_{t \in \mathcal{D}_{\text{calib}}}$ and test scores $(s_t)_{t \in [T]}$ are exchangeable¹ (in particular, if the corresponding pairs of covariates–outcomes are exchangeable), the above procedure guarantees that for all $\alpha \in [0, 1]$,

$$\forall t \in [T], \quad \mathbb{P}(y_t \in C_{1-\alpha}(x_t)) \geq 1 - \alpha. \quad (1)$$

Actually, up to some minor assumption of calibration scores being all distinct, the above probability lies in the interval between $1 - \alpha$ and $1 - \alpha + 1/(T_{\text{calib}} + 1)$, that is, is approximately equal to $1 - \alpha$.

Remark 2 *The literature typically sets $s_{(T_{\text{calib}}+1)} = +\infty$, which is fine to study validity but complicates efficiency analyses (Zaffran et al., 2022). We rather assume that there exists a known constant L such that, for all $t \in \mathcal{D}_{\text{calib}} \cup [T]$, one has $\mathbb{P}(|y_t - \hat{\mu}(x_t)| < L) = 1$. In practical applications, Zaffran et al. (2022) suggest to set $L = 2 \max\{s_t : t \in \mathcal{D}_{\text{calib}}\}$.*

2.2 Online Setting of Conformal Inference: Adaptive Conformal Inference [ACI]

In the online setting, no stochastic assumption is made on the calibration and test scores, and the learner can adapt the choice of prediction intervals sequentially based on past observations.

A natural extension of the setting of conformal inference with these two characteristics was introduced by Gibbs and Candès (2021) under the name of adaptive conformal inference. For theoretical purposes, we consider the simplest online setting of Gibbs and Candès (2021),

1. Random variables $U_1, \dots, U_n$ are said exchangeable if for all permutations $\sigma : [n] \rightarrow [n]$, the n -tuples $(U_1, \dots, U_n)$ and $(U_{\sigma(1)}, \dots, U_{\sigma(n)})$ have the same distribution.

and assume that the training set and calibration set are fixed, exactly as in SCP (see Section 2.1).

At each round, these fixed sets induce a family of nested prediction intervals of the form

$$C_{1-\alpha}(\cdot) = \left[\hat{\mu}(\cdot) \pm s_{\left(\lceil (T_{\text{calib}}+1)(1-\alpha) \rceil\right)} \right]. \quad (2)$$

Then, test data $(x_t, y_t)_{t \in [T]}$ is handled in an online fashion: at round $t \in [T]$, the covariates x_t are revealed, the learner picks some miscoverage rate $\alpha_t \in [0, 1]$ based on past information and outputs the conformal set $C_{1-\alpha_t}(x_t)$. After that, the outcome y_t is revealed and the learner observes whether $y_t \in C_{1-\alpha_t}(x_t)$ or $y_t \notin C_{1-\alpha_t}(x_t)$.

Remark 3 *More complex settings of online conformal prediction exist, where the training and calibration sets, and thus the form of the conformal intervals, are updated over time, which is preferred in practice; see Gibbs and Candès (2024).*

No assumption is made on the generating mechanism of test data $(x_t, y_t)_{t \in [T]}$. It can even be thought of as being picked by a malicious adversary. The probabilistic guarantee (1) holding for each test point is thus replaced by an average guarantee over all test points, as stated next.

The primary objective of the learner is to ensure coverage for a fraction about $1 - \alpha$ of the test data points—a criterion called *validity*:

$$\text{ensure that } \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{y_t \in C_{1-\alpha_t}(x_t)} \quad \text{is larger than about } 1 - \alpha. \quad (3)$$

In this article, we will also be interested in a secondary objective consisting of keeping the intervals $C_{1-\alpha_t}(x_t)$ as small as possible on average—a criterion called *efficiency*. This will of course call for a validity criteria of the form “about $1 - \alpha$ ” rather than just “larger than about $1 - \alpha$ ” in (3).

In a nutshell, in the version of online conformal prediction described above, the learner picks a sequence of data-dependent miscoverage levels $\alpha_t \in [0, 1]$ to ensure approximate coverage on average. This setting is now commonly referred to as adaptive conformal inference [ACI], although the term originally denotes the specific algorithm of Gibbs and Candès (2021, restated in Example 1 below).

2.3 Adaptive Conformal Inference [ACI] as a Repeated Finite Game

Our first main result is a game-theoretic interpretation of ACI as a repeated finite game. To do so, we first formulate ACI as a repeated game with continuous action sets and then construct a reformulation thereof as a repeated game with finite action sets.

Direct formulation of ACI as a game with continuous action sets. We first see ACI as a game where, at stage $t \geq 1$, the learner picks a prediction set $C_{1-\alpha_t}(x_t)$, or alternatively, a miscoverage level $\alpha_t \in [0, 1]$, while the opponent picks the observation $y_t \in \mathbb{R}$. In terms of validity, the question is whether $y_t \in C_{1-\alpha_t}(x_t)$ or $y_t \notin C_{1-\alpha_t}(x_t)$. In terms of efficiency, we evaluate the prediction set through its length $\mathfrak{L}(C_{1-\alpha_t}(x_t))$.

Remark 4 *Note that by construction, the length is independent of the covariate $x_t \in \mathcal{X}$, which is why we will simply write $\mathfrak{L}(C_{1-\alpha_t})$ in the sequel.*

Two helpful properties to reduce action sets to finite sets. We next state two properties that are helpful to reformulate this game with continuous action sets into a finite game.

We first show that the discrete nature of the possible lengths of the prediction sets, dictated by the scores in the calibration set (fixed throughout), entails that the learner actually picks miscoverage levels a_t in the following discrete set $\mathcal{A}$, instead of general miscoverage levels $\alpha_t \in [0, 1]$:

$$\mathcal{A} = \left\{ \frac{k}{T_{\text{calib}} + 1} : k \in \llbracket 0, T_{\text{calib}} + 1 \rrbracket \right\}.$$

Indeed, we define the rounding function $r : [0, 1] \rightarrow \mathcal{A}$ of $[0, 1]$ onto the finite grid $\mathcal{A}$ as

$$r : \alpha \in [0, 1] \mapsto 1 - \frac{\lceil (T_{\text{calib}} + 1)(1 - \alpha) \rceil}{T_{\text{calib}} + 1},$$

i.e., $r(\alpha)$ is the largest value inferior than or equal to α in $\mathcal{A}$. The following property shows that we can assume, with no loss of generality, namely, up to replacing α_t by $a_t = r(\alpha_t)$, that the learner picks miscoverage levels $a_t \in \mathcal{A}$. The property holds by the construction (2) of prediction intervals based on order statistics of non-conformity scores.

Property 5 *For all $\alpha \in [0, 1]$, for all $x \in \mathcal{X}$, we have $C_{1-\alpha}(x) = C_{1-r(\alpha)}(x)$.*

Second, we show that it is equivalent for the opponent to choose an outcome $y_t \in \mathbb{R}$ or some action b_t in a finite set $\mathcal{B}$ to be specified. Indeed, the question is whether the outcome y_t picked by the opponent belongs, or not, to $C_{1-a_t}(x_t)$. The property stated in the next lemma, introduced by Gibbs and Candès (2024), and slightly reworked here, shows that given $\hat{\mu}(x_t)$ and the historical data, the observation y_t can be replaced by an action $b_t \in \mathcal{B}$, where

$$\mathcal{B} = \left\{ \frac{k}{T_{\text{calib}} + 1} : k \in \llbracket 1, T_{\text{calib}} + 1 \rrbracket \right\} = \mathcal{A} \setminus \{0\}.$$

Note that $\mathcal{B}$ does not contain the value 0, unlike $\mathcal{A}$.

Lemma 6 (from Gibbs and Candès, 2024, with a slight adaptation) *The following equalities define a scalar $b_t \in \mathcal{B}$, thus $b_t > 0$:*

$$b_t = \sup\{\beta \in [0, 1] : y_t \in C_{1-\beta}(x_t)\} = \min\{\beta \in \mathcal{B} : y_t \notin C_{1-\beta}(x_t)\}.$$

As a consequence, $\mathbf{1}_{y_t \notin C_{1-a_t}(x_t)} = \mathbf{1}_{b_t \leq a_t}$.

Proof Fix $t \geq 1$ and define b_t as the supremum of the set $\{\beta \in [0, 1] : y_t \in C_{1-\beta}(x_t)\}$, which is non-empty since $y_t \in C_1(x_t)$ by Fact 1. That conformal prediction sets are nested ensures that

$$b_t = \sup\{\beta \in [0, 1] : y_t \in C_{1-\beta}(x_t)\} = \inf\{\beta \in [0, 1] : y_t \notin C_{1-\beta}(x_t)\}, \quad (4)$$

and we show next that the infimum is a minimum.

First, if $b_t = 1$, then $b_t \in \mathcal{B}$; as $y_t \notin C_{1-1}(x_t) = \emptyset$, the infimum in (4) is a minimum, at the value $\beta = 1$.

Second, if $b_t < 1$, then there exists $\beta \in (b_t, 1]$ such that $r(\beta) = r(b_t)$. On the one hand, by Property 5, we have $C_{1-\beta}(x_t) = C_{1-b_t}(x_t)$. On the other hand, since $\beta > b_t$ and by definition of b_t as a supremum, $y_t \notin C_{1-\beta}(x_t)$. Thus, $y_t \notin C_{1-b_t}(x_t)$, which proves that the infimum in (4) is a minimum, at the value $\beta = b_t$.

Finally, we show that $b_t \in \mathcal{B}$. Note that $b_t > 0$ since $y_t \notin C_{1-b_t}(x_t)$ whereas $y_t \in C_1(x_t)$. On the one hand, the characterization of $b_t > 0$ as an infimum implies that $y_t \in C_{1-\beta}(x_t)$ for all $0 \leq \beta < b_t$. On the other hand, the rounding function is such that $0 \leq r(b_t) \leq b_t$ and $C_{1-b_t}(x_t) = C_{1-r(b_t)}(x_t)$. Therefore, using again that $y_t \notin C_{1-b_t}(x_t)$, we obtain $b_t = r(b_t)$, and thus $b_t \in \mathcal{B}$, since $\mathcal{B} = r([0, 1]) \setminus \{0\}$.

The consequence stated at the end of the lemma follows from the characterization of b_t as a minimum. $\blacksquare$

The lemma above allows to rewrite the update scheme of the ACI algorithm from Gibbs and Candès (2021) in terms of the player's and opponent's actions as follows.

Example 1 *The update scheme of the ACI algorithm from Gibbs and Candès (2021) can be written as*

$$\alpha_{t+1} = \alpha_t - \gamma(\mathbf{1}_{b_t \leq a_t} - \alpha) \quad \text{and} \quad a_{t+1} = \begin{cases} r(\alpha_{t+1}), & \text{if } \alpha_{t+1} \in [0, 1], \\ \min\{1, \max\{\alpha_{t+1}, 0\}\}, & \text{otherwise,} \end{cases} \quad (5)$$

where α_{t+1} serves as an intermediate value for choosing $a_{t+1} \in \mathcal{A}$, and $\gamma > 0$ is a fixed step size.

Reformulation of ACI as a repeated finite game. Based on all elements above, we now reformulate ACI as the following repeated finite game. Note that the reward obtained is formed by a pair in $\{0, 1\} \times [0, 2L]$, as we are interested in both validity and efficiency guarantees.

Theorem 7 (first main result) *The setting of adaptive conformal inference [ACI] may be formulated as a repeated finite game, where at each round $t \geq 1$, the learner picks $a_t \in \mathcal{A}$, the opponent picks $b_t \in \mathcal{B}$, the learner receives the reward $m(a_t, b_t)$ given by the vector-valued payoff function $m : \mathcal{A} \times \mathcal{B} \rightarrow \{0, 1\} \times [0, 2L]$ defined by*

$$m : (a, b) \in \mathcal{A} \times \mathcal{B} \mapsto m(a, b) = \left(\mathbf{1}_{b \leq a}, \mathfrak{L}(C_{1-a}) \right). \quad (6)$$

2.4 Links Between the Batch and Online Settings (Between SCP and ACI)

The above formalism is suited for adversarial sequences of test observations $(x_t, y_t)_{t \in [T]}$ but it also encompasses the exchangeable setting: the latter corresponds to an opponent player picking actions b_t uniformly at random in $\mathcal{B}$, as stated next. We denote by $\mathbf{q}_{\text{unif}}$ the uniform distribution on $\mathcal{B}$:

$$\mathbf{q}_{\text{unif}} = \frac{1}{T_{\text{calib}} + 1} \sum_{b \in \mathcal{B}} \delta_b.$$

Lemma 8 *If the calibration scores $(s_k)_{k \in \mathcal{D}_{\text{calib}}}$ and the test score s_t are exchangeable and almost-surely distinct, then $b_t \sim \mathbf{q}_{\text{unif}}$.*

The proof of Lemma 8 relies on the following result, not requiring any stochastic assumption.

Lemma 9 *When the calibration scores $(s_k)_{k \in \mathcal{D}_{\text{calib}}}$ are all distinct, the scalar b_t defined in Lemma 6 satisfies*

$$\forall b \in \mathcal{B}, \quad \{b_t = b\} = \left\{ s_{((T_{\text{calib}}+1)(1-b))} < s_t \leq s_{((T_{\text{calib}}+1)(1-b)+1)} \right\}.$$

Proof The assumption of distinct values of the calibration scores is used to avoid ties: it entails that for all $b \in \mathcal{B}$,

$$\{y_t \notin C_{1-b}(x_t)\} = \left\{ s_t > s_{((T_{\text{calib}}+1)(1-b))} \right\}.$$

Thus, by the definition of b_t as some minimum in Lemma 6 (and since 0 does not belong to $\mathcal{B}$), for all $b \in \mathcal{B}$,

$$\begin{aligned} \{b_t = b\} &= \{y_t \notin C_{1-b}(x_t)\} \setminus \{y_t \notin C_{1-b+1/(T_{\text{calib}}+1)}(x_t)\} \\ &= \left\{ s_{((T_{\text{calib}}+1)(1-b))} < s_t \leq s_{((T_{\text{calib}}+1)(1-b)+1)} \right\}. \end{aligned} \quad \blacksquare$$

We are now able to prove Lemma 8.

Proof of Lemma 8 By exchangeability and the assumption of almost-surely distinct scores, s_t is equally likely to fall into any of the $T_{\text{calib}} + 1$ intervals defined by the scores computed on the calibration set:

$$\forall k \in \llbracket 0, T_{\text{calib}} \rrbracket, \quad \mathbb{P}(s_t \in (s_{(k)}, s_{(k+1)}]) = \mathbb{P}(s_t \in (s_{(k)}, s_{(k+1)})) = \frac{1}{T_{\text{calib}} + 1}.$$

Now, Lemma 9 exactly ensures that each event $\{b_t = b\}$, where $b \in \mathcal{B}$, is of the form of the events considered above, thus these $T_{\text{calib}} + 1$ events are equally likely. $\blacksquare$

3 Second Contribution: Efficiency under a Validity Constraint

We now leverage the reduction of ACI to the repeated finite game of Theorem 7 involving payoffs in terms of coverage and length of conformal intervals.

Our aims are twofold: $(\star)$ achieve validity i.e., $y_t \in C_{1-a_t}(x_t)$ at least $100(1 - \alpha)\%$ of the time on average and in the limit, as stated earlier in (3); and $(\star\star)$ produce efficient prediction sets, i.e., prediction sets with small lengths, on average and in the limit.

Formal statement of the objectives. These two objectives are now formally stated based on a target length ℓ_α as

$$\begin{aligned} \text{Validity:} \quad & \text{ensure} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{y_t \notin C_{1-a_t}(x_t)} = \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{b_t \leq a_t} \leq \alpha; \quad (\star) \\ \text{Efficiency:} \quad & \text{ensure} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathcal{L}(C_{1-a_t}) \leq \ell_\alpha. \quad (\star\star) \end{aligned}$$

The validity objective $(\star)$ is expressed in terms of an empirical miscoverage smaller than α , which corresponds to an empirical coverage larger than $1 - \alpha$.

We strive for a target length ℓ_α as small as possible. This target will be smaller when stronger stochastic assumptions are made on the feature-outcome pairs (or equivalently, on the scores).

Reformulation of these objectives as an approachability problem. With the vector-valued payoff function introduced in Theorem 7, we may reformulate the objectives $(\star)$ and $(\star\star)$ as approaching the closed convex set $[0, \alpha] \times [0, \ell_\alpha]$, i.e.,

Validity and Efficiency:

$$\text{ensure} \quad \frac{1}{T} \sum_{t=1}^T \left(\mathbf{1}_{b_t \leq a_t} \right) \longrightarrow [0, \alpha] \times [0, \ell_\alpha] \quad \text{as } T \rightarrow \infty, \quad (\star \& \star\star)$$

in the sense that $d(\bar{m}_T, [0, \alpha] \times [0, \ell_\alpha]) \rightarrow 0$ as $T \rightarrow \infty$, where

$$\bar{m}_T \stackrel{\text{def}}{=} \frac{1}{T} \sum_{t=1}^T m(a_t, b_t) \stackrel{\text{def}}{=} \frac{1}{T} \sum_{t=1}^T \left(\mathbf{1}_{b_t \leq a_t} \right).$$

Convergences of the above form are called approachability results and were first studied by Blackwell (1956), see Appendix B.1.

The question at hand, and a tool to answer it. The question at hand is therefore what target lengths ℓ_α may be achieved depending on the data-generation mechanisms considered. We first discuss two extreme of such mechanisms: (i) the favorable exchangeable setting of Definition 10, and (ii) the worst-case adversarial setting of Definition 11. We next consider a general framework based on the notion of *Q-restricted opponent's play*, introduced by Bernstein et al. (2014) and recalled in Definition 12, which encompasses practically relevant intermediate settings.

Bernstein et al. (2014) provide a theory called opportunistic approachability to identify target sets that are approachable under the considered *Q*-restrictions; this theory will be the key tool used to answer the question of setting the target lengths ℓ_α .

3.1 Several Degrees of Stochasticity for the Opponent

We first define the two extreme data-generating mechanisms. The assumption of exchangeability is a typical assumption in the batch conformal prediction setting of Section 2.1. It entails that the actions b_t of the opponent in the associated repeated finite game of Theorem 7 are uniformly distributed.

Definition 10 (exchangeable setting) *The exchangeable setting corresponds to the fact that at each round $t \geq 1$, calibration scores $(s_k)_{k \in \mathcal{D}_{\text{calib}}}$ and the test score s_t are exchangeable and almost-surely distinct. By Lemma 8, we have $b_t \sim \mathbf{q}_{\text{unif}}$.*

The adversarial setting refers to not assuming anything on the test observations, which, in turn, corresponds to not assuming anything on the actions b_t in the associated repeated finite game of Theorem 7.

Definition 11 (adversarial setting) *The adversarial setting corresponds to the fact that at each round $t \geq 1$, the opponent freely selects the test observations (x_t, y_t) , possibly based on the outcomes of past rounds (with the only constraint that the resulting test score s_t lies in $[0, L]$). Therefore, nothing can be assumed on the test score $s_t \in [0, L]$ or on the opponent's action b_t .*

A framework interpolating the previous two settings was introduced by Bernstein et al. (2014), in the form of Q -restricted opponent's play. In the definition stated next, $\mathbf{q}_t$ denotes the distribution of the outcome b_t in the associated repeated finite game of Theorem 7. This distribution may depend on the results of the past rounds: $\mathbf{q}_t$ is a predictable random variable.

Definition 12 (setting of Q -restricted opponent's play) *Fix a subset $Q \subseteq \Delta(\mathcal{B})$. Let $\mathbf{q}_t$ denote the distribution of b_t at round $t \geq 1$, given the results of past rounds. We say that an opponent's play is Q -restricted whenever it satisfies*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.}$$

Note that Q is generally unknown to the learner, and that the Cesàro-mean convergence of Definition 12 is a weaker condition than the convergence of $d(\mathbf{q}_t, Q)$ to zero (and, in particular, the fact that $\mathbf{q}_t \in Q$ for all $t \geq 1$).

The exchangeable setting and the adversarial setting are two extreme cases thereof of Definition 12, see Examples 2 and 3 below.

Example 2 *In the exchangeable setting, the opponent's play is, in particular, $\{\mathbf{q}_{\text{unif}}\}$ -restricted, where $\mathbf{q}_{\text{unif}}$ is the uniform distribution on $\mathcal{B}$ (see Lemma 8). Actually, $\{\mathbf{q}_{\text{unif}}\}$ -restriction may also be achieved when the exchangeability assumption between the calibration scores and the test scores s_t is violated for a sublinear number of rounds t .*

Example 3 *The adversarial setting imposes no condition on the opponent, which corresponds to a $\Delta(\mathcal{B})$ -restricted player.*

Example 3 above explains why the assertion that the opponent's play is Q -restricted for some unknown $Q \subseteq \Delta(\mathcal{B})$ is not a real assumption: this is always the case. Sometimes (e.g., in Example 2), this set Q is much smaller than the set $\Delta(\mathcal{B})$ of all distributions. Section 5 will provide further examples of realistic small sets Q .

3.2 Informal Statements of Validity–Efficiency Results Achieved

We state informally weaker results than the ones achieved in Section 4: here, convergences of $\bar{m}_T$, i.e., empirical average miscoverages and lengths, to sets of the form $[0, \alpha] \times [0, \ell_{\alpha, Q}]$, instead of the more precise convergences to subsets $\mathcal{C}_{\alpha, Q}$ thereof as shown in Section 4.

Achievable average length under a Q -restriction. Given a distribution $\mathbf{q} = (q_b)_{b \in \mathcal{B}}$ in $\Delta(\mathcal{B})$, we denote by $\text{mscv}(\alpha, \mathbf{q})$ the largest action in $\mathcal{A}$ satisfying, on average over $\mathbf{q}$, the validity condition, i.e., with miscoverage at most α :

$$\text{mscv}(\alpha, \mathbf{q}) = \max \left\{ a \in \mathcal{A} : \sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq a} \leq \alpha \right\}. \quad (7)$$

Target length: worst-case length under Q . Conceptually, a natural target would be the worst-case length for a valid prediction set associated with a mixed action in Q , namely,

$$\max_{\mathbf{q} \in Q} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}).$$

However, potential discontinuities of m^* on the boundary of Q may arise, which is why, for technical reasons, the target considered is rather (see Lemma 16)

$$\ell_{\alpha, Q} = \lim_{\eta \downarrow 0} \max_{\mathbf{q} : d(\mathbf{q}, Q) \leq \eta} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}),$$

where the maximum is taken over all distributions $\mathbf{q} \in \Delta(\mathcal{B})$ in an η -neighborhood of Q . When there is no ambiguity, we suppress the statement $\mathbf{q} \in \Delta(\mathcal{B})$ in the maximum, as it is understood throughout.

Section 4 will establish formally approachability results that are stronger than convergence to $[0, \alpha] \times [0, \ell_{\alpha, Q}]$ (the convergence will be to a smaller target denoted by $\mathcal{C}_{\alpha, Q}$); but for now, we state informally the former convergence, together with its implications in various scenarios.

Informal Result 13 (second main series of results) *The BO-ACI strategy defined in Section 4, tuned with $\alpha \in (0, 1)$, ensures that for all convex subsets $Q \subseteq \Delta(\mathcal{B})$, for all Q -restricted opponents in the sense of Definition 12,*

$$\bar{m}_T = \frac{1}{T} \sum_{t=1}^T \left(\mathfrak{L}(C_{1-a_t}) \right) \longrightarrow [0, \alpha] \times [0, \ell_{\alpha, Q}] \quad \text{a.s.,} \quad \text{as } T \rightarrow \infty;$$

put differently, this strategy satisfies the validity $(\star)$ and efficiency $(\star\star)$ objectives, with a target length $\ell_{\alpha, Q}$ whenever the opponent is Q -restricted.

This target length equals, for instance,

- $\ell_{\alpha, \{\mathbf{q}_{\text{unif}}\}} = \mathfrak{L}(C_{1-\alpha})$ when the opponent is $\{\mathbf{q}_{\text{unif}}\}$ -restricted and when $\alpha \notin \mathcal{A}$, which is, in particular, the case when data is exchangeable (see Example 2);
- $\mathfrak{L}(C_{1-\alpha+1/(T_{\text{calib}}+1)})$ when data is almost exchangeable and possibly comes with dependencies, with a level of non-exchangeability that is sufficiently small;
- $\mathfrak{L}(C_{1-a^{\mathcal{S}, \mathcal{V}}})$ in the single shift distribution setting, when test scores are i.i.d. according to a distribution $\mathcal{V}$ while calibration scores were i.i.d. according to a distribution $\mathcal{S}$, and where $\mathbf{q}^{\mathcal{S}, \mathcal{V}}$ denotes the distribution induced on the b_t and $a^{\mathcal{S}, \mathcal{V}} = \text{mscv}(\alpha, \mathbf{q}^{\mathcal{S}, \mathcal{V}})$ is the largest element $a \in \mathcal{A}$ such that C_{1-a} satisfies the $1 - \alpha$ validity condition on average under $\mathbf{q}^{\mathcal{S}, \mathcal{V}}$;
- $\ell_{\alpha, \Delta(\mathcal{B})} = 2L$ in the adversarial setting of Definition 11 and Example 3.

Remark 14 *We insist again that the considered strategy is unaware of the value of Q and even of the existence of a potential Q -restriction, but turns out to exploit it if it holds. This justifies the name of “opportunistic approachability” for the underlying theory used (see details in Appendix B.2).*

If instead Q were known to the learner, one may want to adapt the classic, primal, approachability strategy to directly approach $\mathcal{C}_Q$, and not only $\mathcal{C}$: we perform this in Appendix C for the sake of completeness, and also discuss the strategy thus obtained in the exchangeable case, where $Q = \{\mathbf{q}_{\text{unif}}\}$.

Outline of the rest of the article. Section 4 proves the first part of Informal Result 13, namely, the general convergence result. Section 5 provides concrete and realistic scenarios of Q -restrictions, together with the computation of the associated target length $\ell_{\alpha,Q}$; put differently, Section 5 goes over the second part of Informal Result 13.

4 Application of Approachability Theory under Q -Restricted Play (a.k.a. Opportunistic Approachability)

This section is devoted to formally state and prove the first part of Informal Result (13), consisting of the general convergence result.

We resort to the general theory of approachability under Q -restricted play from Bernstein et al. (2014), also known as opportunistic approachability, which is recalled in Appendix B.2. In this section, we restate specifically for ACI the results that can be achieved (and provide in appendix short and direct proofs thereof, tailored to the specific setting considered and bypassing some technical considerations needed for the general theory by Bernstein et al., 2014). To do so, we define in Section 4.1 the target set $\mathcal{C}_{\alpha,Q}$ to be approached, we provide in Section 4.2 some reminder on calibrated forecasters, which are used as a subroutine for the main strategy, introduced and studied in Section 4.3.

4.1 Definition of Target Sets $\mathcal{C}_{\alpha,Q}$ Tailored to Q -Restricted Opponents

The set $\mathcal{C}_\alpha = [0, \alpha] \times [0, 2L]$ is always approachable. The starting point is to note that we may always ensure validity $(\star)$, e.g., by alternating between $a_t = 0$ a fraction $1 - \alpha$ of the time and $a_t = 1$ a fraction α of the time. Given the boundedness assumptions issued (see Remark 2), setting $\ell_\alpha = 2L$ corresponds to ignoring the efficiency constraint $(\star\star)$. Thus, the target set $\mathcal{C}_\alpha = [0, \alpha] \times [0, 2L]$ can always be approached in the sense of $(\star \& \star\star)$.

More ambitious target sets $\mathcal{C}_{\alpha,Q}$ based on Q -restrictions. Now, the point of opportunistic approachability is to design a general strategy of the learner not only approaching a given approachable set, like $\mathcal{C}_\alpha$, but actually approaching a smaller target set $\mathcal{C}_{\alpha,Q} \subseteq \mathcal{C}_\alpha$ whenever the opponent is Q -restricted, with $Q \subseteq \Delta(\mathcal{B})$ being unknown to the learner.

More formally, we consider the bilinear extension of the vector payoff function m defined in Theorem 7: for all probability distributions $\mathbf{p} = (p_a)_{a \in \mathcal{A}} \in \Delta(\mathcal{A})$ and $\mathbf{q} = (q_b)_{b \in \mathcal{B}} \in \Delta(\mathcal{B})$,

$$m(\mathbf{p}, \mathbf{q}) = \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} p_a q_b m(a, b) = \left(\sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} p_a q_b \mathbf{1}_{b \leq a} \right) \left(\sum_{a \in \mathcal{A}} p_a \mathfrak{L}(C_{1-a}) \right).$$

This extended payoff function m and the target set $\mathcal{C}_\alpha = [0, \alpha] \times [0, 2L]$ are such that

$$\forall \mathbf{q} \in \Delta(\mathcal{B}), \quad \exists \mathbf{p}^*(\mathbf{q}) \in \Delta(\mathcal{A}) \quad \text{s.t.} \quad m(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) \in \mathcal{C}_\alpha; \quad (8)$$

indeed, and for instance, a natural choice for $p^*(\mathbf{q})$ is the largest pure action in $\mathcal{A}$ satisfying, on average over $\mathbf{q}$, the validity condition with miscoverage [mscv] at most α :

$$p^*(\mathbf{q}) = \delta_{\text{mscv}(\alpha, \mathbf{q})}, \quad \text{where} \quad \text{mscv}(\alpha, \mathbf{q}) = \max \left\{ a \in \mathcal{A} : \sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq a} \leq \alpha \right\}. \quad (9)$$

Other choices are possible for $p^*(\mathbf{q})$ but we fix the above one.

Equation (9) defines a so-called response function $p^* : \Delta(\mathcal{B}) \rightarrow \Delta(\mathcal{A})$, only returning Dirac masses, from which, based on (8), we may in turn construct the goal function

$$m^* : \mathbf{q} \in \Delta(\mathcal{B}) \mapsto m^*(\mathbf{q}) \stackrel{\text{def}}{=} m(p^*(\mathbf{q}), \mathbf{q}) = \begin{pmatrix} \sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq \text{mscv}(\alpha, \mathbf{q})} \\ \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}) \end{pmatrix} \in \mathcal{C}_\alpha = [0, \alpha] \times [0, 2L]. \quad (10)$$

We exploit this choice of m^* to define smaller target sets $\mathcal{C}_{\alpha, Q}$ for Q -restricted opponents. Namely, $\mathcal{C}_{\alpha, Q}$ is based on the image of Q under the goal function m^* , with some additional twists required for some technical reasons: taking closures of convex hulls and considering the non-increasing limit of images of η -neighborhoods of Q .

Definition 15 (closed convex image under m^*) *The closed convex image of a convex subset $Q \subseteq \Delta(\mathcal{B})$ under the goal function m^* is defined as*

$$\mathcal{C}_{\alpha, Q} = \bigcap_{\eta > 0} \text{cl} \left(\text{conv} \left(\{m^*(\mathbf{q}) : d(\mathbf{q}, Q) \leq \eta\} \right) \right),$$

and satisfies $\mathcal{C}_{\alpha, Q} \subseteq \mathcal{C}_\alpha$.

The approachability result of $(\star \& \star \star)$ will actually be in terms of a convergence to $\mathcal{C}_{\alpha, Q}$. This will be sharper than a mere convergence to the target set $[0, \alpha] \times [0, \ell_{\alpha, Q}]$ initially considered in $(\star \& \star \star)$, as follows from the lemma stated next.

Lemma 16 *The closed convex image of a convex subset $Q \subseteq \Delta(\mathcal{B})$ under the goal function m^* defined in (10) satisfies*

$$\mathcal{C}_{\alpha, Q} \subseteq [0, \alpha] \times [0, \ell_{\alpha, Q}], \quad \text{where} \quad \ell_{\alpha, Q} = \lim_{\eta \downarrow 0} \max_{\mathbf{q} : d(\mathbf{q}, Q) \leq \eta} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}).$$

Proof We argue component by component: we denote by $\Pi_j : (u_1, u_2) \mapsto u_j$ the extraction of the j -th component of a vector $(u_1, u_2) \in \mathbb{R}^2$, where $j \in \{1, 2\}$.

For the first components, Equation (10) states that $\Pi_1(m^*(\mathbf{q})) \in [0, \alpha]$ for all $\mathbf{q} \in \Delta(\mathcal{B})$; since $[0, \alpha]$ is a closed convex set, the definition of $\mathcal{C}_{\alpha, Q}$ entails that $\Pi_1(\mathcal{C}_{\alpha, Q}) \subseteq [0, \alpha]$.

For the second components, fix first $\eta > 0$. Recall that $\Pi_2(m^*(\mathbf{q})) = \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})})$ for all $\mathbf{q} \in \Delta(\mathcal{B})$, where $\mathfrak{L}(C_{1-\text{mscv}(\alpha, \cdot)})$ takes finitely many values. We thus have, for all $\mathbf{q} \in \Delta(\mathcal{B})$ such that $d(\mathbf{q}, Q) \leq \eta$,

$$\Pi_2(m^*(\mathbf{q})) \in \left[0, \max_{\mathbf{q}' : d(\mathbf{q}', Q) \leq \eta} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q}')} \right].$$

The latter interval is a closed convex set, and $\mathcal{C}_{\alpha, Q} \subseteq \text{cl}\left(\text{conv}\left(\{m^*(\mathbf{q}) : d(\mathbf{q}, Q) \leq \eta\}\right)\right)$, so that

$$\Pi_2(\mathcal{C}_{\alpha, Q}) \subseteq \left[0, \max_{\mathbf{q}' : d(\mathbf{q}', Q) \leq \eta} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q}')})\right].$$

The proof is concluded by taking the non-increasing limit as $\eta \downarrow 0$. ■

4.2 Reminder on Calibration

The main strategy, described in Section 4.3, relies on a subroutine consisting in a calibrated forecaster. This subsection therefore provides a reminder on calibration. The setting and learning protocol for calibration are described next, for the finite set $\mathcal{B}$ of Section 2.3.

Setting of calibration and related objectives. At each round $t \geq 1$, the forecaster and the opponent simultaneously pick a mixed action $\mathbf{z}_t \in \Delta(\mathcal{B})$ and a pure action $b_t \in \mathcal{B}$, respectively, where b_t is picked possibly independently at random based on a mixed action denoted by $\mathbf{q}_t \in \Delta(\mathcal{B})$. The mixed actions $\mathbf{z}_t$ and $\mathbf{q}_t$ may depend on the information gathered in the previous rounds. At the end of the round, the forecaster observes b_t .

The goal of the forecaster is to produce calibrated forecasts, in the following sense. For interpretations of these definitions and constructive proofs of existence, we refer the reader, e.g., to Mannor and Stoltz (2010), whose approach relies on Blackwell's approachability theorem, and to the references cited therein (above all, Foster and Vohra, 1998, Fudenberg and Levine, 1999, Foster, 1999).

Definition 17 (calibrated forecaster) *A forecaster is said to be calibrated if, for every strategy of the opponent, the forecasts $(\mathbf{z}_t)_{t \geq 1}$ output satisfy:*

$$\text{for all Borel-measurable sets } G \subseteq \Delta(\mathcal{B}), \quad \lim_{T \rightarrow \infty} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in G} (\mathbf{z}_t - \delta_{b_t}) \right\| = 0 \quad \text{a.s.}$$

There exist calibrated forecasters. They are typically constructed based on a sequence of ε -calibrated forecasters, which we define in Section 6.

The following lemma, stated by Bernstein et al. (2014, Lemma 6) in the case of a calibrated forecaster, guarantees that the Q -restriction property of the mixed actions $\mathbf{q}_t$ taken by the opponent transfers to calibrated forecasts $\mathbf{z}_t$ thereof.

Lemma 18 *Consider any convex subset $Q \subseteq \Delta(\mathcal{B})$, and any Q -restricted opponent. If a forecaster is calibrated, the forecasts $(\mathbf{z}_t)_{t \geq 1}$ output satisfy, almost surely:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) = 0.$$

For the sake of completeness, a proof of the lemma above (and of a similar result for ε -calibrated forecasters) may be found in Appendix A.1.

Algorithm 1 Blackwell Opportunistic Adaptive Conformal Inference [BO-ACI]

Inputs: miscoverage level α ; auxiliary sequential forecaster $\mathcal{F}$ with outputs in $\Delta(\mathcal{B})$

Initialization: play $a_1 = \text{mscv}(\alpha, \mathbf{q}_{\text{unif}})$

At time steps $t = 2, 3, \dots$:

1. Get from $\mathcal{F}$ a probabilistic forecast $\mathbf{z}_t \in \Delta(\mathcal{B})$ of the opponent's pure action
 2. Observe x_t , play $a_t = \text{mscv}(\alpha, \mathbf{z}_t)$, and thus output the prediction set $C_{1-a_t}(x_t)$
 3. Observe y_t , deduce the corresponding b_t from Lemma 6, and feed $\mathcal{F}$ with b_t
-

4.3 Strategy BO-ACI: Blackwell Opportunistic Adaptive Conformal Inference

In this section, we present our algorithm, referred to as Blackwell Opportunistic Adaptive Conformal Inference [BO-ACI]. The strategy relies on an auxiliary forecaster $\mathcal{F}$, which at each round t , outputs, possibly at random, a probabilistic forecast $\mathbf{z}_t \in \Delta(\mathcal{B})$ of the result $b_t \in \mathcal{B}$ of the opponent's pick for (x_t, y_t) . Based on this forecast, the learner then deterministically selects the pure action $a_t \in \mathcal{A}$ such that $p^*(\mathbf{z}_t) = \delta_{a_t}$, i.e., $a_t = \text{mscv}(\alpha, \mathbf{z}_t)$, and finally outputs the prediction set $C_{1-a_t}(x_t)$.

The description above is summarized in Algorithm 1.

It is a standard trick in sequential learning to use a calibrated auxiliary forecaster $\mathcal{F}$ and optimally respond to the probabilistic forecasts output (through p^* in our strategy). This approach was first introduced by Perchet (2009) in the context of Blackwell approachability, and was later incorporated as a central component of the general construction of opportunistically approaching strategies by Bernstein et al. (2014).

The BO-ACI strategy of Algorithm 1 is a simple instance of these more general strategies. In Appendix A we propose a simple, direct, and self-contained proof of its performance, as stated next in Theorem 19, exploiting the specificities of adaptive conformal inference.

Theorem 19 (Opportunistic approachability for adaptive conformal inference)

The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with $\alpha \in (0, 1)$, ensures that for all convex subsets $Q \subseteq \Delta(\mathcal{B})$, for all Q -restricted opponents in the sense of Definition 12,

$$\overline{m}_T = \frac{1}{T} \sum_{t=1}^T \left(\mathbf{1}_{b_t \leq a_t} \right) \rightarrow \mathcal{C}_{\alpha, Q} \quad a.s., \quad \text{as } T \rightarrow \infty.$$

Theorem 19, together with Lemma 16, proves the first part of Informal Result 13. Theorem 19 is proved in Appendix A.2: we provide therein a concise, carefully simplified and revisited, self-contained, proof (by adapting the proof by Bernstein et al., 2014 to the specific application considered).

5 Examples of Target Lengths $\ell_{\alpha, Q}$ Achieved Depending on the Degree of Stochasticity Q

This section is devoted to formally define the cases stated in the second part of Informal Result (13) and review which target lengths $\ell_{\alpha, Q}$ may be achieved in each case. We will

not consider the same order as stated therein and will rather proceed in increasing order of complexity of the modeling or of numbers of concepts needed.

The various corollaries stated in this section are obtained by considering particular choices of Q in Theorem 19, each corresponding to one of the informal results presented earlier in the article. These specializations illustrate how the general theorem recovers the different regimes of interest within a unified framework.

A useful tool. We first state a corollary of Theorem 19 in the case of $Q = \{\mathbf{q}\}$ being a singleton: the average payoff vectors $\bar{m}_T$ obtained by the BO-ACI strategy then converge to a point whenever $m^\star$ is continuous at $\mathbf{q}$.

Corollary 20 *Consider $\alpha \in (0, 1)$ and $\mathbf{q} \in \Delta(\mathcal{B})$ such that $m^\star$ is continuous at $\mathbf{q}$. The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$, is such that, in particular, if the opponent is $\{\mathbf{q}\}$ -restricted, i.e.,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \|\mathbf{q}_t - \mathbf{q}\| = 0, \quad \text{then} \quad \lim_{T \rightarrow \infty} \bar{m}_T = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left(\mathbf{1}_{b_t \leq a_t} \right) = m^\star(\mathbf{q}) \quad \text{a.s.}$$

Proof We apply Theorem 19 with $Q = \{\mathbf{q}\}$; it suffices to note that

$$\mathcal{C}_{\alpha, \{\mathbf{q}\}} = \bigcap_{\eta > 0} \text{cl} \left(\text{conv} \left(\{m^\star(\mathbf{q}') : d(\mathbf{q}', \mathbf{q}) \leq \eta\} \right) \right) = \{m^\star(\mathbf{q})\}$$

by continuity of $m^\star$ at $\mathbf{q}$. ■

5.1 Exchangeable Case

The case of exchangeable data (see Definition 10) is encompassed in the case of a $\{\mathbf{q}_{\text{unif}}\}$ -restricted opponent, see Example 2. We deal with the latter, more general, case and show in the course of the proof of Corollary 22 that the average vector payoffs $\bar{m}_T$ converge to the set $\mathcal{C}_{\alpha, \{\mathbf{q}_{\text{unif}}\}}$, and that the latter admits a closed-form expression as a singleton:

$$\mathcal{C}_{\alpha, \{\mathbf{q}_{\text{unif}}\}} = \left\{ (r(\alpha), \mathfrak{L}(C_{1-\alpha}))^\top \right\}. \quad (11)$$

The miscoverage level is only $r(\alpha) \leq \alpha$, for technical reasons due to rounding.

Before we do so, and for the sake of comparison to the literature, we introduce the concept of general strategies: strategies that guarantee validity in all settings. It of course suffices to ask this requirement in the adversarial setting.

Definition 21 (general strategies) *A strategy for outputting conformal prediction sets is call general at miscoverage level α if it ensures validity at this level in the adversarial setting: for all sequences of data $(x_t, y_t)_{t \geq 1}$ picked by an opponent,*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{y_t \notin C_{1-\alpha_t}(x_t)} \leq \alpha.$$

By Theorem 19, together with Lemma 16 and the consideration of $Q = \Delta(\mathcal{B})$, the BO-ACI strategy of Algorithm 1 is a general strategy, which turns out to also guarantee the desirable objective in the exchangeable setting given by controlling $(\star\star)$ by the length of $C_{1-\alpha}$. This is stated next in Corollary 22.

To the best of our knowledge, the BO-ACI strategy of Algorithm 1 is the first strategy to simultaneously achieve these two properties. In particular, no general strategy considered in the literature is known to also satisfy the efficiency guarantee $\mathfrak{L}(C_{1-\alpha})$ in the exchangeable setting. In fact, the existing analysis of the ACI algorithm (of Gibbs and Candès, 2021, restated in Example 1) illustrates the opposite phenomenon: Zaffran et al. (2022) showed that it yields a strictly larger empirical average length than $\mathfrak{L}(C_{1-\alpha})$ in the exchangeable setting. The additional length depends linearly on the step size γ used in the ACI algorithm, on the nominal miscoverage level α , and on the underlying distribution of the calibration scores. They interpret the result as indicating that the ACI algorithm of Gibbs and Candès (2021) adapts to a potentially adversarial opponent but at the cost of a larger average length in the exchangeable setting. (We note however that the proof of this result relies on a simplifying assumption introduced to bypass technical difficulties arising from the finiteness of the calibration set, and therefore considers a slightly different setting from ours.)

Corollary 22 (exchangeable setting) *Consider $\alpha \notin \mathcal{A}$. The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with α , is a general strategy at miscoverage level α (see Definition 21) and it also satisfies, against a $\{\mathbf{q}_{\text{unif}}\}$ -restricted opponent, and thus, in the exchangeable setting, the following efficiency guarantee:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathfrak{L}(C_{1-a_t}) = \mathfrak{L}(C_{1-\alpha}) \quad \text{a.s.} \quad (12)$$

The proof of Corollary 22 is located in Appendix A.4; it consists of showing that the continuity required in Corollary 20 is always satisfied for $\alpha \notin \mathcal{A}$, and of establishing the closed-form expression (11).

Remark 23 *If $\alpha \in \mathcal{A}$, then the asymptotic efficiency guarantee in (12) deteriorates to $\mathfrak{L}(C_{1-\alpha+1/(T_{\text{calib}}+1)})$ as an upper bound, due to a potential discontinuity jump. We obtain this result as a special case of Corollary 28 stated later in this article; see the comments after its statement.*

5.2 Some Intermediate Scenarios

The exchangeability considered in the previous section is a relaxation of i.i.d. data in the direction of lack of independence. We now study a different relaxation, in the direction of lack of identical distribution, under independence assumptions. More precisely, for all three scenarios considered at the beginning of each subsection, we issue the following stochastic assumptions.

Common stochastic assumptions. We assume that the calibration scores are drawn i.i.d. from some (unknown) diffuse distribution $\mathcal{S}$ with support contained in $[0, L]$. At time $t \geq 1$, the test score s_t is drawn independently at random according to some (unknown)

distribution $\mathcal{S}_t$ that may depend on the previous observations, and with support contained in $[0, L]$. To emphasize the dependence on the distributions, we introduce the notation $\mathbf{q}_t = \mathbf{q}^{\mathcal{S}, \mathcal{S}_t}$ for the distribution of the opponent's action b_t at time $t \geq 1$. In particular, given that $\mathcal{S}$ is diffuse and that scores are therefore almost-surely distinct, $\mathbf{q}^{\mathcal{S}, \mathcal{S}} = \mathbf{q}_{\text{unif}}$ is the uniform distribution on $\mathcal{B}$.

Relaxation of these stochastic assumptions. The stochastic assumptions above are only needed to motivate the associated Q -restrictions of the opponent's play that will be stated in Assumptions 24 and 27. The latter will cover more general cases, not requiring strict independence assumptions.

5.2.1 SINGLE DISTRIBUTIONAL SHIFT

This section studies, in particular, a setting of single distributional shift. We assume, in this setting, that $\mathcal{S}_t = \mathcal{V}$ for all $t \geq 1$ for some constant distribution $\mathcal{V}$, i.e., conditionally on the training and calibration data, the test scores are i.i.d. according to $\mathcal{V}$. This setting covers both the fully i.i.d. case where $\mathcal{V} = \mathcal{S}$ and situations where the test distribution $\mathcal{V}$ differs from the calibration distribution $\mathcal{S}$ but is stationary. It also includes cases where calibration data may have been inadvertently incorporated into the training process, thereby altering the effective distribution underlying subsequent observations.

We may generalize this setting through the following assumption.

Assumption 24 *The opponent's play is $\{\mathbf{q}^{\mathcal{S}, \mathcal{V}}\}$ -restricted.*

Under this assumption, the desirable efficiency bound is $\mathfrak{L}(C_{1-a^{\mathcal{S}, \mathcal{V}}})$, the length of the interval associated with $a^{\mathcal{S}, \mathcal{V}} = \text{mscv}(\alpha, \mathbf{q}^{\mathcal{S}, \mathcal{V}})$, the largest element $a \in \mathcal{A}$ such that C_{1-a} satisfies the $1 - \alpha$ validity condition on average under $\mathbf{q}^{\mathcal{S}, \mathcal{V}}$. The counterpart to (and extension of) Corollary 22 under this assumption is the following.

Corollary 25 (single distributional shift) *Consider $\alpha \in (0, 1)$. The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with α , is a general strategy at miscoverage level α (see Definition 21) and it also satisfies, against a $\{\mathbf{q}^{\mathcal{S}, \mathcal{V}}\}$ -restricted opponent, and thus, in single distributional shift setting described above, the following efficiency guarantee: if m^* is continuous at $\mathbf{q}^{\mathcal{S}, \mathcal{V}}$,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathfrak{L}(C_{1-a_t}) = \mathfrak{L}(C_{1-a^{\mathcal{S}, \mathcal{V}}}) \quad \text{a.s.} \quad (13)$$

The first part of Corollary 25 (that BO-ACI is a general strategy) was already stated right after Definition 21. The second part is a mere application of Corollary 20, thanks to the explicit continuity assumption (which was unnecessary in Corollary 22).

5.2.2 ALMOST-EXCHANGEABLE SETTING

Our motivating example, taking place in the stochastic setting described at the beginning of Section 5.2 corresponds to the case where the distributions $\mathcal{S}_t$ may differ slightly from $\mathcal{S}$ and change over time, but with controlled variations, in some sense to be made precise. To describe it with the required detail, we first establish a useful connection in terms of distances between mixed actions in $\Delta(\mathcal{B})$ and distributions of scores.

Lemma 26 *In the stochastic setting described at the beginning of Section 5.2, for any $t \geq 1$, we have that*

$$\left\| \mathbf{q}_{\text{unif}} - \mathbf{q}^{\mathcal{S}, \mathcal{S}_t} \right\|_1 \leq 2 \, \text{d}_{\text{TV}}(\mathcal{S}, \mathcal{S}_t) \quad \text{a.s.}$$

Proof The test score s_t is an independent draw from $\mathcal{S}_t$ conditionally on the training, calibration, and past test observations. Because $\mathcal{S}$ is diffuse, Lemma 9 is applicable and ensures that for all $b \in \mathcal{B}$,

$$q_b^{\mathcal{S}, \mathcal{S}_t} = \mathbb{P}_{s_t \sim \mathcal{S}_t}(b_t = b) = \mathbb{P}_{s_t \sim \mathcal{S}_t}\left(s_{((T_{\text{calib}}+1)(1-b))} < s_t \leq s_{((T_{\text{calib}}+1)(1-b)+1)}\right) \quad \text{a.s.}$$

The same argument applied to $s \sim \mathcal{S}$ implies that

$$\left\| \mathbf{q}^{\mathcal{S}, \mathcal{S}} - \mathbf{q}^{\mathcal{S}, \mathcal{S}_t} \right\|_1 = \sum_{i=0}^{T_{\text{calib}}} \left| \mathbb{P}_{s \sim \mathcal{S}}\left(s \in (s_{(i)}, s_{(i+1)}]\right) - \mathbb{P}_{s_t \sim \mathcal{S}_t}\left(s_t \in (s_{(i)}, s_{(i+1)}]\right) \right| \quad \text{a.s.}$$

Therefore, the definition of the total variation distance, together with classic properties of histograms applied to the (almost-surely distinct) order statistics $s_{(i)}$, gives that

$$\left\| \mathbf{q}^{\mathcal{S}, \mathcal{S}} - \mathbf{q}^{\mathcal{S}, \mathcal{S}_t} \right\|_1 \leq 2 \, \text{d}_{\text{TV}}(\mathcal{S}, \mathcal{S}_t) \quad \text{a.s.}$$

The proof is concluded by recalling that $\mathbf{q}^{\mathcal{S}, \mathcal{S}} = \mathbf{q}_{\text{unif}}$. ■

We can now define the ε -exchangeable setting (or, by abuse, the almost-exchangeable setting) in terms of a convergence condition on the total variation distance between the distribution $\mathcal{S}$ of the calibration scores and the distributions $(\mathcal{S}_t)_{t \geq 1}$ of the test scores. This setting notably encompasses the case of independent but non-identically distributed test scores, provided that the total variation distances $\text{d}_{\text{TV}}(\mathcal{S}, \mathcal{S}_t)$ are small (inferior to some threshold $\varepsilon/2$) except for a sublinear number of rounds t .

Example 4 *The ε -exchangeable setting, where $\varepsilon \in [0, 2]$, corresponds to the case where*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{\text{d}_{\text{TV}}(\mathcal{S}, \mathcal{S}_t) > \varepsilon/2\}} = 0 \quad \text{a.s.} \quad (14)$$

In this case, the opponent's play is Q_ε -restricted, where Q_ε is the ℓ_1 -ball in $\Delta(\mathcal{B})$ centered at $\mathbf{q}_{\text{unif}}$ and with radius ε , namely,

$$Q_\varepsilon = \left\{ \mathbf{q} \in \Delta(\mathcal{B}) : \left\| \mathbf{q} - \mathbf{q}_{\text{unif}} \right\|_1 \leq \varepsilon \right\}. \quad (15)$$

Proof Indeed, by definition of Q_ε , we have that

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q_\varepsilon) &= \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{\|\mathbf{q}_t - \mathbf{q}_{\text{unif}}\|_1 > \varepsilon\}} (\|\mathbf{q}_t - \mathbf{q}_{\text{unif}}\|_1 - \varepsilon) \\ &\leq \frac{2}{T} \sum_{t=1}^T \mathbf{1}_{\{\text{d}_{\text{TV}}(\mathcal{S}, \mathcal{S}_t) > \varepsilon/2\}} \quad \text{a.s.,} \end{aligned}$$

where the inequality follows from the fact that ℓ_1 -distances between distributions are smaller than 2, together with Lemma 26 (to bound the indicator function by another indicator function). The desired convergence to 0 follows from the assumption (14). $\blacksquare$

Based on the previous example, we will more generally consider the following assumption in the rest of this section.

Assumption 27 *The opponent's play is Q_ε -restricted for some $\varepsilon \in [0, 2]$, where Q_ε is defined in (15).*

We state next the associated efficiency guarantee, in Corollary 28. Its proof is also located in Appendix A.4.

Corollary 28 (ε -exchangeable setting) *Consider $\alpha \in [1/(T_{\text{calib}} + 1), 1)$. The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with α , is a general strategy at miscoverage level α (see Definition 21) and it also satisfies, against a Q_ε -restricted opponent, where $\varepsilon \in [0, 1/(T_{\text{calib}} + 1))$, and thus, in the ε -exchangeable setting of Example 4, the following efficiency guarantee:*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathfrak{L}(C_{1-a_t}) \leq \mathfrak{L}(C_{1-\alpha+1/(T_{\text{calib}}+1)}) \quad \text{a.s.}$$

Comparison to the exchangeable case and discussion. A comparison of Corollaries 22 and 28 reveals that the mild price to pay for not facing fully exchangeable scores is a replacement of the ideal length $\mathfrak{L}(C_{1-\alpha})$ by the smallest possible strictly larger value $\mathfrak{L}(C_{1-\alpha+1/(T_{\text{calib}}+1)})$, providing that the departure from exchangeability is small enough.

Note that Corollary 28 also encompasses the result of Remark 23, i.e., $\alpha \in \mathcal{A}$ in the exchangeable setting. Indeed, the opponent's play is in particular Q_ε -restricted for any $\varepsilon > 0$ in the exchangeable setting.

Also, the specific case of $\alpha < 1/(T_{\text{calib}} + 1)$ is not covered by Corollary 28, but is actually pathological and coincides with the result stated in Corollary 22 for $\alpha = 0$.

5.2.3 ALMOST-EXCHANGEABLE SETTING, CONTINUED: WITH EXPLICIT DEPENDENCIES

In this section, we show that similar results as in the previous motivating example of Section 5.2.2 can be obtained under a typical dependence structure, which notably encompasses some cases of time series. In particular, we show that Assumption 27 is also satisfied.

We consider again the stochastic setting described at the beginning of Section 5.2 and recall that the distributions $\mathcal{S}_t$ may depend on previous observations. Before formalizing this final, more complex, motivating example, we need to establish a similar relationship as in Lemma 26 but with respect to a distance between distributions that is more adapted to the dependent setting, namely, the Kolmogorov distance d_K . The latter is defined as the supremum norm between cumulative distribution functions.

Lemma 29 *In the stochastic setting described at the beginning of Section 5.2, for any $t \geq 1$, we have that*

$$\left\| \mathbf{q}_{\text{unif}} - \mathbf{q}^{\mathcal{S}, \mathcal{S}_t} \right\|_1 \leq 2(T_{\text{calib}} + 1) d_K(\mathcal{S}, \mathcal{S}_t) \quad \text{a.s.,}$$

where d_K denotes the Kolmogorov distance between distributions.

Proof Recall that, under the same assumptions, we showed in the proof of Lemma 26 that

$$\left\| \mathbf{q}_{\text{unif}} - \mathbf{q}^{\mathcal{S}, \mathcal{S}_t} \right\|_1 = \sum_{i=0}^{T_{\text{calib}}} \left| \mathbb{P}_{s \sim \mathcal{S}}(s \in (s_{(i)}, s_{(i+1)}]) - \mathbb{P}_{s_t \sim \mathcal{S}_t}(s_t \in (s_{(i)}, s_{(i+1)}]) \right| \quad \text{a.s.}$$

Therefore, several triangular inequalities, the fact that the supports of $\mathcal{S}$ and $\mathcal{S}_t$ are both included in $[0, L] = [s_{(0)}, s_{(T_{\text{calib}}+1)}]$, and the definition of the Kolmogorov distance as the supremum norm between the cumulative distribution functions, entail that

$$\left\| \mathbf{q}_{\text{unif}} - \mathbf{q}^{\mathcal{S}, \mathcal{S}_t} \right\|_1 \leq 2 \sum_{i=0}^{T_{\text{calib}}} \left| \mathbb{P}_{s \sim \mathcal{S}}(s \leq s_{(i)}) - \mathbb{P}_{s_t \sim \mathcal{S}_t}(s_t \leq s_{(i)}) \right| \leq 2(T_{\text{calib}} + 1) d_K(\mathcal{S}, \mathcal{S}_t) \quad \text{a.s.},$$

which is the claimed bound. $\blacksquare$

By the exact same arguments as for the ε -exchangeable setting of Section 5.2.2, we get that the opponent's play is Q_ε -restricted provided that the following condition holds:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\{d_K(\mathcal{S}, \mathcal{S}_t) > \varepsilon / (2T_{\text{calib}} + 2)\}} = 0 \quad \text{a.s.} \quad (16)$$

We now exhibit a natural dependence structure under which such a control on the Kolmogorov distances arises. Following Zaffran et al. (2022), we consider the case of test scores being absolute values of a clipped auto-regressive process of order 1. This modeling is classic in time series forecasting, where forecast errors $r_t = y_t - \hat{\mu}(x_t)$, also known as residuals, are often temporally correlated. Such a dependence may stem either from non-stationarity in the data-generating process or from model misspecification, when the predictor $\hat{\mu}$ fails to capture all relevant dynamics. Although not formally equivalent, it is natural to assume that the calibration scores, which are i.i.d. according to $\mathcal{S}$, are the absolute value of underlying calibration residuals that are i.i.d. with distribution Ξ , supported on $[-L, L]$. In line with this perspective, we model the series of test residuals as an auto-regressive process (e.g., of order 1) with i.i.d. innovations drawn from Ξ .

Example 5 Let $(\xi_t)_{t \geq 1}$ be an i.i.d. innovation process with distribution Ξ (the one of the calibration residuals), and let $\phi > 0$ be a (typically small) auto-regressive parameter. We consider the setting where the test scores are absolute values of the clipped AR(1) process $(r_t)_{t \geq 1}$ initialized at $r_1 = \xi_1$ and recursively defined by

$$\forall t \geq 2, \quad r_t = \max\{\min\{\phi r_{t-1} + \xi_t, L\}, -L\}, \quad \text{i.e.,} \quad s_t = |r_t|.$$

We denote by $\mathcal{S}_t$ the distribution of s_t conditional to past observations (training data and previous residual r_{t-1}). Note that the support of $\mathcal{S}_t$ is included in $[0, L]$.

If we further assume that the density of $\mathcal{S}$ is upper bounded by some constant $\|f_{\mathcal{S}}\|_\infty$, then we get the following bound on the Kolmogorov distances:

$$\forall t \geq 1, \quad d_K(\mathcal{S}, \mathcal{S}_t) \leq L\phi \|f_{\mathcal{S}}\|_\infty \quad \text{a.s.}$$

Proof First, the construction of the scores, a triangular inequality, and the fact that the support of $\mathcal{S}_{t-1}$ is included in $[0, L]$ entail that

$$s_t \leq |\phi r_{t-1} + \xi_t| \leq |\xi_t| + L\phi \quad \text{a.s.}$$

Second, either $|\phi r_{t-1} + \xi_t| \leq L$ or $|\phi r_{t-1} + \xi_t| > L$; by a reverse triangular inequality in the first case, and the fact that $s_t = L$ while $|\xi_t| \in [0, L]$ in the second case, we get that

$$|\xi_t| - L\phi \leq s_t \quad \text{a.s.}$$

Therefore, we have that $|\xi_t| - L\phi \leq s_t \leq |\xi_t| + L\phi$ almost surely; hence, since $|\xi_t| \sim \mathcal{S}$, we obtain

$$\forall u \in [0, L], \quad \mathbb{P}_{s \sim \mathcal{S}}(s \leq u - L\phi) \leq \mathbb{P}_{s_t \sim \mathcal{S}_t}(s_t \leq u) \leq \mathbb{P}_{s \sim \mathcal{S}}(s \leq u + L\phi) \quad \text{a.s.}$$

Based on the above inequalities, we can now control the Kolmogorov distance between $\mathcal{S}$ and $\mathcal{S}_t$ by substituting the cumulative distribution function of $\mathcal{S}_t$ by the cumulative distribution function of $\mathcal{S}$ evaluated at two close points, namely,

$$\begin{aligned} d_K(\mathcal{S}, \mathcal{S}_t) &\stackrel{\text{def}}{=} \sup_{u \in [0, L]} \max \left\{ \mathbb{P}_{s_t \sim \mathcal{S}_t}(s_t \leq u) - \mathbb{P}_{s \sim \mathcal{S}}(s \leq u), \mathbb{P}_{s \sim \mathcal{S}}(s \leq u) - \mathbb{P}_{s_t \sim \mathcal{S}_t}(s_t \leq u) \right\} \\ &\leq \sup_{u \in [0, L]} \max \left\{ \mathbb{P}_{s \sim \mathcal{S}}(s \in (u, u + L\phi]), \mathbb{P}_{s \sim \mathcal{S}}(s \in (u - L\phi, u]) \right\} \quad \text{a.s.} \end{aligned}$$

Finally, the assumption on the density of $\mathcal{S}$, together with the above inequality, entails that

$$d_K(\mathcal{S}, \mathcal{S}_t) \leq L\phi \|f_{\mathcal{S}}\|_{\infty} \quad \text{a.s.} \quad \blacksquare$$

In the dependent setting depicted in Example 5, we showed that the Kolmogorov distances are controlled in the sense of (16), with $\varepsilon = L\phi \|f_{\mathcal{S}}\|_{\infty}$. This quantity is small under weak dependence (small ϕ) and for a non-concentrated distribution $\mathcal{S}$ (small $\|f_{\mathcal{S}}\|_{\infty}$). Consequently, the opponent's play is Q_{ε} -restricted by Lemma 29, so that Assumption 27 and Corollary 28 are applicable.

5.3 On Efficiency Objectives in the Adversarial Setting

This section is devoted to the adversarial setting of Definition 11 and Example 3, which corresponds to $Q = \Delta(\mathcal{B})$, i.e., no restriction on the opponent's play.

Point of this section. We set p^* and m^* in a simple way in (8) and (9) and picked, in particular, a non-randomized response function p^* outputting Dirac masses.

These definitions are such that there exist distributions $\mathbf{q} \in \Delta(\mathcal{B})$ with $\text{mscv}(\alpha, \mathbf{q}) = 0$ and thus $m^*(\mathbf{q}) = (0, 2L)^{\top}$.

There also exist other distributions $\mathbf{q} \in \Delta(\mathcal{B})$ with $\text{mscv}(\alpha, \mathbf{q}) = 1/(T_{\text{calib}} + 1)$ and $m^*(\mathbf{q}) = (\alpha, 2s_{(T_{\text{calib}})})^{\top}$. The latter quantity could be close to $(\alpha, 2L)^{\top}$ so that, with the simple definitions considered, Theorem 19 does not seem to guarantee anything more specific than a convergence of $\overline{m}_T$ to $[0, \alpha] \times [0, 2L]$ in the adversarial setting.

Yet, we have to acknowledge that convergence to $[0, \alpha] \times [0, (1 - \alpha)2L]$ may be achieved, with a randomized p^* , and that this convergence is the best achievable one. Both results

are formally stated below in Lemma 30. This lemma highlights a fundamental limitation of the fully adversarial setting: while validity can always be satisfied, efficiency cannot simultaneously be controlled against any opponent in a non-trivial way, i.e., by a bound strictly smaller than $(1 - \alpha)2L$.

However, for the sake of a simpler exposition, we rather considered in Section 4.1 the target set $\mathcal{C}_\alpha = [0, \alpha] \times [0, 2L]$, i.e., we neglected the $1 - \alpha$ term in the efficiency component. The set $\mathcal{C}_\alpha$ is also approachable by Lemma 30 (and with the same simple strategy). Remark 31 hints at the complications that would arise from considering the smaller target set $[0, \alpha] \times [0, (1 - \alpha)2L]$. We therefore preferred to obtain slightly suboptimal efficiency results in the adversarial case rather than picking a more complex p^* than in (9), at the cost of additional technicalities in the analysis.

Lemma 30 *The target set $[0, \alpha] \times \{(1 - \alpha)2L\}$ and thus the target set $[0, \alpha] \times [0, (1 - \alpha)2L]$ are approachable by the learner; put differently, there exists a (simple) strategy of the learner such that, the following inequalities hold against any opponent, almost surely:*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{b_t \leq a_t} \leq \alpha \quad \text{and} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathfrak{L}(C_{1-a_t}) \leq (1 - \alpha)2L.$$

Conversely, for all $\varepsilon > 0$, the target set $[0, \alpha] \times [0, (1 - \alpha)2L - \varepsilon]$ is not approachable by the learner; put differently, the inequalities above cannot hold almost surely against any opponent when the second condition is replaced by $\leq (1 - \alpha)2L - \varepsilon$.

Lemma 30 basically indicates that in a fully adversarial setting, only trivial results may be achieved when it comes to efficiency. This is why we restricted the behavior of the opponent, in a way already considered by the literature of online learning, that encompasses both exchangeable and adversarial settings as extreme cases; namely, through Q -restrictions, see Section 3.1.

Proof For the first statement of the lemma, consider the following strategy² for the learner: at each round $t \geq 1$, pick $a_t = 0$ with probability $1 - \alpha$ and $a_t = 1$ with probability α , i.e., play $\mathbf{p}_t = (1 - \alpha)\delta_0 + \alpha\delta_1$. By the boundedness assumption on the scores, $y_t \in C_1(x_t)$ with probability 1. Also, by design, the sets $C_0(x_t)$ and $C_1(x_t)$ have respective lengths 0 and $2L$. Thus, $m(\mathbf{p}_t, b_t) \in [0, \alpha] \times \{(1 - \alpha)2L\}$ for all round $t \geq 1$. By martingale convergence (e.g., the Hoeffding-Azuma inequality together with the Borel-Cantelli lemma),

$$\limsup_{T \rightarrow \infty} \left| \frac{1}{T} \sum_{t=1}^T m(a_t, b_t) - \underbrace{\frac{1}{T} \sum_{t=1}^T m(\mathbf{p}_t, b_t)}_{\in [0, \alpha] \times \{(1 - \alpha)2L\}} \right| = 0 \quad \text{a.s.}$$

which concludes the proof of the first statement.

For the second statement of the lemma, we fix $\varepsilon > 0$ and show that Blackwell's dual condition is not satisfied for the target set $\mathcal{C}^{(\varepsilon)} = [0, \alpha] \times [0, (1 - \alpha)2L - \varepsilon]$, which yields the desired lack of approachability, by Theorem 37 of Appendix B.1. We consider $\mathbf{q} = \delta_{1/(T_{\text{calib}} + 1)}$

2. The associated conformal sets correspond to the typical example of a poor but valid sequence of prediction sets that is used in the literature of conformal inference to emphasize that validity alone is not a sufficiently interesting objective.

and proceed by contradiction: we assume that there exists $\mathbf{p} = (p_a)_{a \in \mathcal{A}} \in \Delta(\mathcal{A})$ such that $m(\mathbf{p}, \mathbf{q}) \in \mathcal{C}^{(\varepsilon)}$. Since $s_{(T_{\text{calib}})} < L$ by the choice of L (see Remark 2), we can consider (x, y) such that $s_{(T_{\text{calib}})} < |y - \hat{\mu}(x)| < L$, e.g., by taking any $x \in \mathcal{X}$ and letting $y = \hat{\mu}(x) + (s_{(T_{\text{calib}})} + L)/2$. This choice of (x, y) indeed corresponds to $\mathbf{q} = \delta_{1/(T_{\text{calib}}+1)}$ and is such that the only conformal set that contains y is $C_1(x)$. Therefore $m(\mathbf{p}, \mathbf{q}) \in \mathcal{C}^{(\varepsilon)}$ implies in particular that the prediction set $C_1(x)$ is selected with probability at least $1 - \alpha$, i.e., $p_0 \geq 1 - \alpha$. In that case, the second component of $m(\mathbf{p}, \mathbf{q})$ is at least $(1 - \alpha) \mathfrak{L}(C_1) = (1 - \alpha)2L$, thus contradicting the fact that $m(\mathbf{p}, \mathbf{q}) \in \mathcal{C}^{(\varepsilon)}$. $\blacksquare$

Remark 31 With $[0, \alpha] \times [0, (1 - \alpha)2L]$ as a target set, a suitable response function should satisfy, for all $\mathbf{q} \in \Delta(\mathcal{B})$,

$$p_{\text{rand}}^*(\mathbf{q}) \in \arg \min \left\{ \sum_{a \in \mathcal{A}} p_a \mathfrak{L}(C_{1-a}) \quad \text{over } \mathbf{p} \in \Delta(\mathcal{A}) \quad \text{subject to} \quad \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} p_a q_b \mathbf{1}_{b \leq \alpha} \leq \alpha \right\}.$$

Such a response function p_{rand}^* would lead to randomized actions, as opposed to the Dirac-mass-valued function p^* considered with the target set $\mathcal{C} = [0, \alpha] \times [0, 2L]$. Our analysis could be adapted to handle this slightly more complex case (the original version of Bernstein et al., 2014 does so) but this would not, in general, bring any substantial improvement in the efficiency guarantees achieved, except perhaps in the adversarial setting, while leading to less elementary proofs in Appendix A.

6 BO-ACI Strategy Based on ε -Calibrated Forecasters

In this section, we extend the theoretical guarantees of the BO-ACI strategy of Algorithm 1 to the case where the auxiliary forecaster is only ε -calibrated as per the following definitions, starting with the notion of disjoint ε -covering.

Definition 32 (disjoint ε -covering) Given $\varepsilon > 0$, a finite partition $\mathcal{P} = \{P_k : k \in [N]\}$ of $\Delta(\mathcal{B})$ into measurable subsets is a disjoint ε -covering if, for every $k \in [N]$, there exists $\mathbf{c}_k \in P_k$ such that

$$P_k \subseteq B(\mathbf{c}_k, \varepsilon) \stackrel{\text{def}}{=} \{\mathbf{q} \in \Delta(\mathcal{B}) : \|\mathbf{c}_k - \mathbf{q}\|_1 \leq \varepsilon\}.$$

We refer to $\{\mathbf{c}_k : k \in [N]\}$ as centers associated with the disjoint ε -covering $\mathcal{P}$.

Definition 33 (ε -calibrated forecaster) Given $\varepsilon > 0$, and a disjoint ε -covering $\mathcal{P}$ with associated centers $\{\mathbf{c}_k : k \in [N]\}$, a forecaster is said to be ε -calibrated if, for every strategy of the opponent, its forecasts $(\mathbf{z}_t)_{t \geq 1}$ take values in the set of centers, and if, denoting for each $t \geq 1$ by $k_t \in [N]$ the index such that $\mathbf{z}_t = \mathbf{c}_{k_t}$, the following holds:

$$\limsup_{T \rightarrow \infty} \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{c}_k - \delta_{b_t}) \right\| \leq \varepsilon \quad \text{a.s.}$$

For technical reasons and with no loss of generality, we assume that the underlying disjoint ε -covering $\mathcal{P} = \{P_k : k \in [N]\}$ of $\Delta(\mathcal{B})$ used by the auxiliary ε -calibrated forecaster $\mathcal{F}$ is compatible with p^* , or equivalently, with $\text{mscv}(\alpha, \cdot)$, in the following sense:

$$\forall k \in [N], \exists a \in \mathcal{A} \text{ s.t. } P_k \subseteq R_a \stackrel{\text{def}}{=} \{\mathbf{q} \in \Delta(\mathcal{B}) : \text{mscv}(\alpha, \mathbf{q}) = a\}, \quad (17)$$

i.e., p^* and $\text{mscv}(\alpha, \cdot)$ take constant values on each P_k .

This compatibility indeed comes with no loss of generality as we may transform any disjoint ε -covering $\mathcal{P}' = \{P'_k : k \in [N']\}$ into

$$\mathcal{P} = \{P'_k \cap R_a : k \in [N'] \text{ and } a \in \mathcal{A} \text{ s.t. } P'_k \cap R_a \neq \emptyset\}$$

to make it compatible with p^* and $\text{mscv}(\alpha, \cdot)$. We critically use here that $\mathcal{A}$ is finite.

We can now state the adaptation of Theorem 19 to an ε -calibrated forecaster with respect to a disjoint ε -covering compatible with p^* . We strengthen the assumption of Q -restriction of the opponent used in Definition 40 and Theorem 19: we do not only assume that $d(\mathbf{q}_t, Q)$ vanishes on Cesàro average but instead that $d(\mathbf{q}_t, Q) = 0$, i.e., $\mathbf{q}_t \in \text{cl}(Q)$, for all $t \geq 1$.

Theorem 34 (Opportunistic ε -approachability for adaptive conformal inference)

Fix $\varepsilon \in (0, 1)$. The BO-ACI strategy of Algorithm 1, when tuned with $\alpha \in (0, 1)$ and run with an auxiliary ε -calibrated forecaster $\mathcal{F}$ relying on a disjoint ε -covering compatible with p^* in the sense of (17), is an ε -opportunistically approaching strategy in the following sense: for any opponent, for any convex subset $Q \subseteq \Delta(\mathcal{B})$,

$$\text{for all } t \geq 1, \quad d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.}$$

$$\text{entails} \quad \limsup_{T \rightarrow \infty} d(\bar{\mathbf{m}}_T, \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}) \leq (2L + 1)(\varepsilon + 2\sqrt{\varepsilon}) \quad \text{a.s.,}$$

$$\text{where } \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}} = \text{cl}\left(\text{conv}\left(\{m^*(\mathbf{q}) : d(\mathbf{q}, Q) \leq \sqrt{\varepsilon}\}\right)\right).$$

7 Discussion and Future Work

In this work, we formulate adaptive conformal inference [ACI] as a repeated two-player game with finite action sets and vector-valued payoffs. This opens a toolbox containing Blackwell approachability and its opportunistic extension, which we leverage to construct a strategy that guarantees validity in all cases at the desired level and adapts the efficiency guarantees to the underlying, unknown, degree of stochasticity of the opponent. This “best of many worlds” property provides a unified perspective on exchangeable, adversarial, and intermediate settings for adaptive conformal inference.

Future work and research perspectives. A first natural extension of this work includes establishing finite-time convergence rates for the validity and efficiency guarantees. The validity criterion, when considered in isolation, can be satisfied at a rate of $1/T$, as proved directly by Gibbs and Candès (2021) and as can be obtained also by the characterization of slow versus fast rates for approachability by Mannor and Perchet (2013). What would be the (optimal) rates when simultaneously considering both criteria, validity and

efficiency? The analysis of the BO-ACI strategy relies, on the one hand, on the use of a calibrated auxiliary forecaster: calibration may be achieved at a rate of $T^{-1/(1+|B|)}$. However, Marinov et al. (2026) recently showed that this calibration subroutine could be bypassed, which improves the associated rates to $T^{-1/3}$ or $T^{-1/4}$, depending on the (lack of) computational efficiency of the approach. The second important element is the convergence stated in Definition 12 of Q -restricted opponents: it should come with a specific rate. Both rates may then be combined to obtain rates on the m -approachability of the sets $\mathcal{C}_{\alpha,Q}$, but it is highly unclear whether one would obtain the best achievable rates doing so, in our specific problem of validity and efficiency in ACI.

More generally, the results presented in this article should be put in perspective with the indicated computationally efficient variant of opportunistic approachability developed by Marinov et al. (2026)³, which builds on recent developments in polynomial-time ε -calibration by Peng (2025) and Fishelson et al. (2026).

Third and finally, it would be interesting to extend our framework to handle sequential updates of the training and calibration sets; this could be performed by considering time-dependent payoff function m_t .

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3. This work was led in parallel to ours; the first version of the present submission was posted in October 2025 (see <https://arxiv.org/abs/2510.15824>) before any version of Marinov et al. (2026), only publicly posted in February 2026.

Outline of the Appendices

The appendices provide first the proofs of all results stated in the main body:

- Appendix A provides elementary proofs of the general convergence result of Section 4, of its instantiation to specific scenarios performed in Section 5, and of the extension to ε -calibration proposed in Section 6.

They then provide reminders on (and occasional extensions of) approachability theory and opportunistic approachability, together with the rederivations of some facts. These sections may of course be skipped by a reader familiar with approachability theory. More specifically,

- Appendix B offers a general exposition of and reminder on approachability theory and its extension by Bernstein et al. (2014) consisting of opportunistic approachability, with mere statements of results and proofs omitted;
- Appendix C goes over Remark 14 and shows that when the set Q of restricted play is known to the learner, opportunistic approachability is not needed and that the classic primal approachability strategy can be adapted to guarantee convergence to C_Q instead of $\mathcal{C}$, with a focus on the exchangeable case $Q = \{\mathbf{q}_{\text{unif}}\}$;
- Appendix D revisits the theory of opportunistic approachability established by Bernstein et al. (2014) for so-called D-sets: it performs two technical twists (one technical correction, consisting of taking closures, while showing the necessity thereof; and one relaxation, consisting of not requiring piecewise continuity of the response function) and provides a simplified and less technical analysis than in the original reference (while keeping its spirit; simplifications take place at the margins).

Appendix A. Proofs of the Main Results, Located in Sections 4–5–6

This appendix proves all the results (lemmas, theorems, corollaries) located in Sections 4–5–6, and does so through elementary and self-contained proofs.

A.1 Proof of Lemma 18 of Section 4.2

The following lemma encompasses and generalizes Lemma 18 of Section 4.2: it indeed covers both the calibrated case of Lemma 18 and an ε -calibrated result that will be useful when proving Theorem 34 of Section 6.

Lemma 35 (Calibration under Q -restricted opponent) *Consider any convex subset $Q \subseteq \Delta(\mathcal{B})$, and any Q -restricted opponent. If a forecaster is calibrated, respectively, ε -calibrated, the forecasts $(\mathbf{z}_t)_{t \geq 1}$ output satisfy, almost surely:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) = 0, \quad \text{respectively,} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) \leq \varepsilon.$$

Lemma 35 was stated and proved, in the calibrated case, by Bernstein et al. (2014, Lemma 6). We provide below an alternative, elementary proof of this result, also covering the case of ε -calibration.

Proof Fix a convex subset $Q \subseteq \Delta(\mathcal{B})$ and a sequence of mixed actions $(\mathbf{q}_t)_{t \geq 1}$ output by a Q -restricted opponent. We decompose the proof in three steps: we show first a general result; second, we instantiate it for an ε -calibrated forecaster; third, we use it repeatedly to get the result for a calibrated forecaster.

Step 1: a general result. We fix $\varepsilon > 0$. We show that, for any finite set $\{\mathbf{u}_k \in \Delta(\mathcal{B}) : k \in [N]\}$, and for all sequences of indices $(k_t)_{t \geq 1}$ picked by the forecaster,

$$\begin{aligned} \limsup_{T \rightarrow \infty} \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{u}_k - \delta_{b_t}) \right\| &\leq \varepsilon \quad \text{a.s.} \\ \text{entails} \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{u}_{k_t}, Q) &\leq \varepsilon \quad \text{a.s.} \end{aligned} \tag{18}$$

By martingale convergence (e.g., repeated applications of the Hoeffding-Azuma inequality together with the Borel-Cantelli lemma), the left-hand side of (18) is equivalent to

$$\limsup_{T \rightarrow \infty} \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{u}_k - \mathbf{q}_t) \right\| \leq \varepsilon \quad \text{a.s.} \tag{19}$$

We now introduce the empirical averages of the opponent's mixed actions over the rounds when k is picked:

$$\bar{\mathbf{q}}_k(T) = \begin{cases} \frac{1}{N_k(T)} \sum_{t=1}^T \mathbf{1}_{k_t=k} \mathbf{q}_t & \text{if } N_k(T) \stackrel{\text{def}}{=} \sum_{t=1}^T \mathbf{1}_{k_t=k} \geq 1, \\ \mathbf{0} & \text{if } N_k(T) = 0. \end{cases}$$

By grouping according to the values of k_t for the first equality, by a triangular inequality for the inequality, and by definition of the $\bar{\mathbf{q}}_k(T)$ for the second equality, the target quantity in (18) can be rewritten as

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T d(\mathbf{u}_{k_t}, Q) &= \sum_{k=1}^N \frac{N_k(T)}{T} d(\mathbf{u}_k, Q) \\ &\leq \sum_{k=1}^N \frac{N_k(T)}{T} \|\mathbf{u}_k - \bar{\mathbf{q}}_k(T)\| + \sum_{k=1}^N \frac{N_k(T)}{T} d(\bar{\mathbf{q}}_k(T), Q) \\ &= \underbrace{\sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{u}_k - \mathbf{q}_t) \right\|}_{\limsup \dots \leq \varepsilon \text{ by (19)}} + \underbrace{\sum_{k=1}^N \frac{N_k(T)}{T} d(\bar{\mathbf{q}}_k(T), Q)}_{\rightarrow 0 \text{ by } Q\text{-restriction}}. \end{aligned}$$

We substituted the limit behavior (19) for the first sum, while the second sum can be seen to vanish, as follows. Indeed, by convexity of Q , the function $d(\cdot, Q)$ is also convex (Rockafellar and Wets, 1998, Example 2.25), and thus, for k such that $N_k(T) \geq 1$,

$$d(\bar{\mathbf{q}}_k(T), Q) = d\left(\frac{1}{N_k(T)} \sum_{t=1}^T \mathbf{1}_{k_t=k} \mathbf{q}_t, Q\right) \leq \frac{1}{N_k(T)} \sum_{t=1}^T \mathbf{1}_{k_t=k} d(\mathbf{q}_t, Q).$$

Therefore,

$$\sum_{k=1}^N \frac{N_k(T)}{T} d(\bar{\mathbf{q}}_k(T), Q) \leq \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q),$$

where the right-hand side vanishes by the assumption of Q -restricted opponent.

Step 2. The stated claim for an ε -calibrated forecaster follows immediately by applying (18) to the set of centers from Definition 33 (i.e., $\mathbf{u}_k = \mathbf{c}_k$).

Step 3. The application to a calibrated forecaster is a slightly more complex; we actually show that its forecasts satisfy:

$$\forall \varepsilon > 0, \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) \leq 2\varepsilon. \quad (20)$$

To that end, fix $\varepsilon > 0$. We consider a disjoint ε -covering $\mathcal{P}_\varepsilon = \{P_k : k \in [N_\varepsilon]\}$ as per Definition 32, with associated centers $\{\mathbf{c}_k : k \in [N_\varepsilon]\}$. We can now define the mapping $\mathbf{z} \mapsto \kappa_\varepsilon(\mathbf{z}) \in [N_\varepsilon]$ given by $\mathbf{z} \in P_{\kappa_\varepsilon(\mathbf{z})}$. (Note that the mapping is well defined because $\mathcal{P}_\varepsilon$ is a partition.) By construction of κ_ε and the definition of $\mathcal{P}_\varepsilon$,

$$\forall t \geq 1, \quad \{\kappa_\varepsilon(\mathbf{z}_t) = k\} = \{\mathbf{z}_t \in P_k\} \subseteq \{\|\mathbf{z}_t - \mathbf{c}_k\|_1 \leq \varepsilon\}. \quad (21)$$

The calibration property of Definition 17, used with $G = P_k$ for a given $k \in [N_\varepsilon]$, ensures that

$$\lim_{T \rightarrow \infty} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\kappa_\varepsilon(\mathbf{z}_t)=k} (\mathbf{z}_t - \delta_{b_t}) \right\| = \lim_{T \rightarrow \infty} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in P_k} (\mathbf{z}_t - \delta_{b_t}) \right\| = 0 \quad \text{a.s.}$$

Summing the above limit over the finitely many $k \in [N_\varepsilon]$, we get

$$\lim_{T \rightarrow \infty} \sum_{k=1}^{N_\varepsilon} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\kappa_\varepsilon(\mathbf{z}_t)=k} (\mathbf{z}_t - \delta_{b_t}) \right\| = 0 \quad \text{a.s.}$$

Repeated applications of the triangular inequality based on (21), and the fact that the ℓ_1 -norm is larger than the Euclidean norm, then ensure that

$$\limsup_{T \rightarrow \infty} \sum_{k=1}^{N_\varepsilon} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\kappa_\varepsilon(\mathbf{z}_t)=k} (\mathbf{c}_k - \delta_{b_t}) \right\| \leq \varepsilon \quad \text{a.s.}$$

The limit above corresponds to the premise of the implication (18); the latter therefore yields that

$$\frac{1}{T} \sum_{t=1}^T d(\mathbf{c}_{\kappa_\varepsilon(\mathbf{z}_t)}, Q) \leq \varepsilon \quad \text{a.s.}$$

Now, repeated applications of the triangular inequality, again based on (21), guarantee that

$$\frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) \leq \frac{1}{T} \sum_{t=1}^T d(\mathbf{c}_{\kappa_\varepsilon(\mathbf{z}_t)}, Q) + \varepsilon,$$

so that property (20) follows by collecting all elements together. $\blacksquare$

A.2 Proof of Theorem 19 of Section 4.3

Theorem 19 (Opportunistic approachability for adaptive conformal inference)

The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with $\alpha \in (0, 1)$, ensures that for all convex subsets $Q \subseteq \Delta(\mathcal{B})$, for all Q -restricted opponents in the sense of Definition 12,

$$\bar{m}_T = \frac{1}{T} \sum_{t=1}^T \left(\mathbf{1}_{b_t \leq a_t} \right) \longrightarrow \mathcal{C}_{\alpha, Q} \quad \text{a.s.,} \quad \text{as } T \rightarrow \infty.$$

This theorem could be obtained as an instance of the general theory of opportunistic approachability by Bernstein et al. (2014), which we recall in Appendix B. However, when focusing on the specific payoff function m , response function $p^\star$ (which is piecewise constant), and goal function $m^\star$, we may bypass many technicalities required to build a general theory. We provide below a concise, carefully simplified and revisited, self-contained, proof.

Proof Fix a convex subset $Q \subseteq \Delta(\mathcal{B})$ and a sequence of mixed actions $(\mathbf{q}_t)_{t \geq 1}$ output by a Q -restricted opponent; we show that $d(\bar{m}_T, \mathcal{C}_{\alpha, Q}) \rightarrow 0$ a.s.

To do so, we consider the following upper bound, based on a triangular inequality:

$$d(\bar{m}_T, \mathcal{C}_{\alpha, Q}) \leq \left\| \bar{m}_T - \frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t) \right\| + d\left(\frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t), \mathcal{C}_{\alpha, Q}\right).$$

Each of the two steps of the proof consists of proving that each of the two summands vanishes almost surely in the limit.

Step 1. First, we recall that the BO-ACI strategy of Algorithm 1 only picks pure actions $a_t \in \mathcal{A}$, given by $a_t = \text{mscv}(\alpha, \mathbf{z}_t)$. Hence, we can define a partition $(R_a)_{a \in \mathcal{A}}$ of $\Delta(\mathcal{B})$ based on the inverse images of $\text{mscv}(\alpha, \cdot)$, namely, $R_a = \{\mathbf{q} \in \Delta(\mathcal{B}) : \text{mscv}(\alpha, \mathbf{q}) = a\}$. The

action a_t picked is such that $\mathbf{z}_t \in R_{a_t}$, and we may group rounds according to the $(R_a)_{a \in \mathcal{A}}$:

$$\frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t) = \sum_{a \in \mathcal{A}} \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} m(a, \mathbf{z}_t) = \sum_{a \in \mathcal{A}} M_a \left(\frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} \mathbf{z}_t \right)$$

and

$$\bar{m}_T = \frac{1}{T} \sum_{t=1}^T m(a_t, b_t) = \sum_{a \in \mathcal{A}} M_a \left(\frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} \delta_{b_t} \right),$$

where the existence of a matrix M_a such that $M_a \mathbf{z} = m(a, \mathbf{z})$ for all $\mathbf{z} \in \Delta(\mathcal{B})$ follows from the linear extension of m . Second, the calibration property of Definition 17, used with $G = R_a$, exactly guarantees that for each $a \in \mathcal{A}$,

$$\lim_{T \rightarrow \infty} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} (\mathbf{z}_t - \delta_{b_t}) \right\| = 0 \quad \text{a.s.}$$

Collecting all elements together, resorting to a triangular inequality, and using the upper bound $\|m(a, b)\| \leq 2L + 1$ for all $(a, b) \in \mathcal{A} \times \mathcal{B}$, we therefore have

$$\left\| \bar{m}_T - \frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t) \right\| \leq \sum_{a \in \mathcal{A}} (2L + 1) \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} (\mathbf{z}_t - \delta_{b_t}) \right\| \longrightarrow 0 \quad \text{a.s.},$$

as announced.

Step 2. We use the definition of a_t as $p^*(\mathbf{z}_t) = \delta_{a_t}$ for the rewriting $m(a_t, \mathbf{z}_t) = m^*(\mathbf{z}_t)$; we show that

$$d\left(\frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t), \mathcal{C}_{\alpha, Q}\right) = d\left(\frac{1}{T} \sum_{t=1}^T m^*(\mathbf{z}_t), \mathcal{C}_{\alpha, Q}\right) \longrightarrow 0 \quad \text{a.s.}$$

Fix $\eta > 0$ for now and introduce the sets of rounds where the forecasts $\mathbf{z}_t$ belong to the η -neighborhood of Q :

$$\forall T \geq 1, \quad \mathcal{V}_T(\eta) = \{t \in [T] : d(\mathbf{z}_t, Q) \leq \eta\}.$$

Their cardinalities $|\mathcal{V}_T(\eta)|$ satisfy $T - |\mathcal{V}_T(\eta)| = o(T)$; indeed, by Markov's inequality, and then by the Q -restriction property of the calibrated forecaster established in Lemma 35,

$$\frac{T - |\mathcal{V}_T(\eta)|}{T} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{d(\mathbf{z}_t, Q) > \eta} \leq \frac{1}{\eta} \left(\frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) \right) \xrightarrow{T \rightarrow \infty} 0 \quad \text{a.s.} \quad (22)$$

A triangular inequality entails that

$$\begin{aligned} & d\left(\frac{1}{T} \sum_{t=1}^T m^*(\mathbf{z}_t), \mathcal{C}_{\alpha, Q}\right) \\ & \leq \underbrace{\left\| \frac{1}{T} \sum_{t=1}^T m^*(\mathbf{z}_t) - \frac{1}{|\mathcal{V}_T(\eta)|} \sum_{t \in \mathcal{V}_T(\eta)} m^*(\mathbf{z}_t) \right\|}_{\leq 2(2L+1)(T-|\mathcal{V}_T(\eta)|)/T} + d\left(\frac{1}{|\mathcal{V}_T(\eta)|} \sum_{t \in \mathcal{V}_T(\eta)} m^*(\mathbf{z}_t), \mathcal{C}_{\alpha, Q}\right), \end{aligned}$$

where the upper bound indicated for the first term is obtained by the $(2L+1)$ -boundedness of m thus of m^* (see Step 1) and the triangle inequality

$$\begin{aligned} & \left\| \frac{1}{T} \sum_{t=1}^T m^*(z_t) - \frac{1}{|\mathcal{V}_T(\eta)|} \sum_{t \in \mathcal{V}_T(\eta)} m^*(z_t) \right\| \\ & \leq \frac{1}{T} \sum_{t \notin \mathcal{V}_T(\eta)} \|m^*(z_t)\| + \left(\frac{1}{|\mathcal{V}_T(\eta)|} - \frac{1}{T} \right) \sum_{t \in \mathcal{V}_T(\eta)} \|m^*(z_t)\| \\ & \leq (2L+1) \left(\frac{T - |\mathcal{V}_T(\eta)|}{T} + \left(\frac{1}{|\mathcal{V}_T(\eta)|} - \frac{1}{T} \right) |\mathcal{V}_T(\eta)| \right) = 2(2L+1) \frac{T - |\mathcal{V}_T(\eta)|}{T}. \end{aligned}$$

We also note that the set $\mathcal{C}_{\alpha,Q}$ is convex as the limit of convex sets. Thus, the function $d(\cdot, \mathcal{C}_{\alpha,Q})$ is also convex (Rockafellar and Wets, 1998, Example 2.25), so that, for all $T \geq 1$,

$$d\left(\frac{1}{|\mathcal{V}_T(\eta)|} \sum_{t \in \mathcal{V}_T(\eta)} m^*(z_t), \mathcal{C}_{\alpha,Q}\right) \leq M_\eta \stackrel{\text{def}}{=} \sup\left\{d(m^*(z), \mathcal{C}_{\alpha,Q}) : z \in \Delta(\mathcal{B}) \text{ s.t. } d(z, Q) \leq \eta\right\}.$$

Collecting all elements, we proved so far that for all $\eta > 0$,

$$\limsup_{T \rightarrow \infty} d\left(\frac{1}{T} \sum_{t=1}^T m^*(z_t), \mathcal{C}_{\alpha,Q}\right) \leq M_\eta \quad \text{a.s.}, \quad (23)$$

and it now suffices to show that $M_\eta \rightarrow 0$ as $\eta \downarrow 0$. The proof for the aforementioned convergence relies on topological arguments, detailed in Appendix A.3 below, which concludes the proof. $\blacksquare$

A.3 An Auxiliary Topological Result

For each $\eta > 0$, we introduce

$$\mathcal{C}_{\alpha,Q}^\eta = \text{cl}\left(\text{conv}\left(\{m^*(\mathbf{q}) : d(\mathbf{q}, Q) \leq \eta\}\right)\right)$$

and note that, by definitions and nestedness of the sets $\mathcal{C}_{\alpha,Q}^\eta$,

$$M_\eta \leq \sup_{\mathbf{c} \in \mathcal{C}_{\alpha,Q}^\eta} d(\mathbf{c}, \mathcal{C}_{\alpha,Q}) \quad \text{and} \quad \mathcal{C}_{\alpha,Q} = \bigcap_{n \geq 1} \mathcal{C}_{\alpha,Q}^{1/n}.$$

In addition, the sets $\mathcal{C}_{\alpha,Q}^{1/n}$ are bounded (since m^* is bounded) and closed thus compact. The proof of $M_{1/n} \rightarrow 0$, which corresponds to $M_\eta \rightarrow 0$ by nestedness, is therefore provided by the lemma below.

Lemma 36 (see, e.g., Henrot and Pierre, 2005, page 30) *A nested sequence of non-empty compact subsets $(K_n)_{n \geq 1}$ of $\mathbb{R}^d$ converges to their intersection $K = \bigcap_{n \geq 1} K_n$ in the Hausdorff distance:*

$$d_H(K_n, K) = \max\left\{\sup_{\mathbf{k} \in K_n} d(\mathbf{k}, K), \sup_{\mathbf{k} \in K} d(\mathbf{k}, K_n)\right\} \xrightarrow{n \rightarrow \infty} 0.$$

Proof As $K \subseteq K_n$ for all $n \geq 1$, we have $d(\mathbf{k}, K_n) = 0$ for all $\mathbf{k} \in K$, thus $\sup_{\mathbf{k} \in K} d_H(\mathbf{k}, K_n) = 0$; we are anyway interested in the other supremum.

By continuity of the point-to-set distance function $d(\cdot, K)$ and by compactness of the non-empty sets K_n , there exists a sequence of points $(\mathbf{k}_n)_{n \geq 1}$ such that $\mathbf{k}_n \in K_n$ and $\sup_{\mathbf{k} \in K_n} d(\mathbf{k}, K) = d(\mathbf{k}_n, K)$ for all $n \geq 1$. By the Bolzano-Weierstrass theorem, we can extract a subsequence $\mathbf{k}_{\varphi(n)}$ converging to some $\mathbf{k}_\infty \in \mathbb{R}^d$, where φ is a non-decreasing application from positive integers to positive integers. The sequence of sets is nested, therefore, first, $(d(\mathbf{k}_n, K))_{n \geq 1}$ is non-increasing thus converging, and second, that limit can only be the one of $d(\mathbf{k}_{\varphi(n)}, K)$, which, by continuity, is $d(\mathbf{k}_\infty, K)$. Now, again by nestedness, for each $n \geq 1$, the points $\mathbf{k}_{\varphi(N)}$, with $N \geq n$, belong to $K_{\varphi(n)}$, which is closed; therefore, their limit $\mathbf{k}_\infty$ also belongs to $K_{\varphi(n)}$. This being true for all $n \geq 1$, and the sets being nested, we have that $\mathbf{k}_\infty \in K$. Collecting all elements, we thus proved

$$d(\mathbf{k}_n, K) \xrightarrow{n \rightarrow \infty} d(\mathbf{k}_\infty, K) = 0,$$

which concludes the proof. ■

A.4 Proof of Corollaries 22 and 28 of Sections 5.1 and 5.2

Corollary 22 (exchangeable setting) *Consider $\alpha \notin \mathcal{A}$. The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with α , is a general strategy at miscoverage level α (see Definition 21) and it also satisfies, against a $\{\mathbf{q}_{\text{unif}}\}$ -restricted opponent, and thus, in the exchangeable setting, the following efficiency guarantee:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathfrak{L}(C_{1-at}) = \mathfrak{L}(C_{1-\alpha}) \quad \text{a.s.} \quad (12)$$

Proof We establish below the continuity of m^* at $\mathbf{q}_{\text{unif}}$ and also provide its value at that point. Both facts, together with Corollary 20, ensure the convergence

$$\lim_{T \rightarrow \infty} \overline{m}_T = m^*(\mathbf{q}_{\text{unif}}) = \left(\begin{array}{c} r(\alpha) \\ \mathfrak{L}(C_{1-\alpha}) \end{array} \right).$$

For the continuity, recall that the response function is defined as $p^* : \mathbf{q} \mapsto \delta_{\text{mscv}(\alpha, \mathbf{q})}$, where the function mscv is defined as

$$\text{mscv}(\alpha, \cdot) : \mathbf{q} \in \Delta(\mathcal{B}) \mapsto \max \left\{ a \in \mathcal{A} : \sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq a} \leq \alpha \right\}.$$

Since $\mathcal{B} = \mathcal{A} \setminus \{0\}$,

$$\forall a \in \mathcal{A}, \quad \sum_{b \in \mathcal{B}} \frac{1}{|\mathcal{B}|} \mathbf{1}_{b \leq a} = a, \quad \text{so that} \quad \text{mscv}(\alpha, \mathbf{q}_{\text{unif}}) = r(\alpha). \quad (24)$$

Take $\varepsilon = \min\{\alpha - r(\alpha), r(\alpha) - \alpha + 1/|\mathcal{B}|\}$: since $\alpha \notin \mathcal{A}$, we have $\varepsilon > 0$. We now show that for all $\mathbf{q} \in \Delta(\mathcal{B})$ with $\|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 < \varepsilon$, we have $\text{mscv}(\alpha, \mathbf{q}) = r(\alpha) = \text{mscv}(\alpha, \mathbf{q}_{\text{unif}})$, which

will ensure the continuity of p^* , thus of m^* given the definition of the latter. For all $\mathbf{q}$ such that $\|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 < \varepsilon$, by a first triangular inequality,

$$\sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq r(\alpha)} < \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathbf{1}_{b \leq r(\alpha)} + \varepsilon = r(\alpha) + \varepsilon \leq \alpha.$$

Similarly, we have by another triangular inequality that

$$\sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq r(\alpha) + 1/|\mathcal{B}|} > \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathbf{1}_{b \leq r(\alpha) + 1/|\mathcal{B}|} - \varepsilon = r(\alpha) + 1/|\mathcal{B}| - \varepsilon \geq \alpha,$$

so that $\text{mscv}(\alpha, \mathbf{q}) = r(\alpha)$, as announced.

The value of $\{m^*(\mathbf{q}_{\text{unif}})\}$ follows: its first component equals $r(\alpha)$ given the left statement of (24), while its second component equals $\mathfrak{L}(C_{1-r(\alpha)}) = \mathfrak{L}(C_{1-\alpha})$. $\blacksquare$

Corollary 28 (ε -exchangeable setting) *Consider $\alpha \in [1/(T_{\text{calib}} + 1), 1)$. The BO-ACI strategy of Algorithm 1, when run with an auxiliary calibrated forecaster $\mathcal{F}$ and tuned with α , is a general strategy at miscoverage level α (see Definition 21) and it also satisfies, against a Q_ε -restricted opponent, where $\varepsilon \in [0, 1/(T_{\text{calib}} + 1))$, and thus, in the ε -exchangeable setting of Example 4, the following efficiency guarantee:*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathfrak{L}(C_{1-a_t}) \leq \mathfrak{L}(C_{1-\alpha+1/(T_{\text{calib}}+1)}) \quad \text{a.s.}$$

Proof We recall that $|\mathcal{B}| = T_{\text{calib}} + 1$ and will rather write $|\mathcal{B}|$ in the proof. Fix α, ε as in the statement, i.e., $\alpha \in [1/|\mathcal{B}|, 1)$ and $\varepsilon \in [0, 1/|\mathcal{B}|)$. We apply Theorem 19 with $Q_\varepsilon = \{\mathbf{q} \in \Delta(\mathcal{B}) : \|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 \leq \varepsilon\}$ and get the convergence

$$\bar{m}_T \longrightarrow \mathcal{C}_{Q_\varepsilon} \quad \text{a.s.},$$

where, by Lemma 16, we have $\mathcal{C}_{Q_\varepsilon} \subseteq [0, \alpha] \times \left[0, \lim_{\eta \downarrow 0} \max_{\mathbf{q}: d(\mathbf{q}, Q_\varepsilon) \leq \eta} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})})\right]$.

We now show that the limit involved in the definition of the above set is smaller than $\mathfrak{L}(C_{1-\alpha+1/|\mathcal{B}|})$, which will conclude the proof. By a triangular inequality together with a standard ℓ_1 -Euclidean norm inequality, we get that, for all $\eta > 0$,

$$\{\mathbf{q} \in \Delta(\mathcal{B}) : d(\mathbf{q}, Q_\varepsilon) \leq \eta\} \subseteq \{\mathbf{q} \in \Delta(\mathcal{B}) : \|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 \leq \varepsilon + \sqrt{|\mathcal{B}|} \eta\}.$$

Recall that $\varepsilon < 1/|\mathcal{B}|$; therefore $\varepsilon + \sqrt{|\mathcal{B}|} \eta < 1/|\mathcal{B}|$ for $\eta > 0$ small enough, and thus

$$\lim_{\eta \downarrow 0} \max_{\mathbf{q}: d(\mathbf{q}, Q_\varepsilon) \leq \eta} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}) \leq \max_{\mathbf{q}: \|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 < 1/|\mathcal{B}|} \mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}).$$

We fix $\mathbf{q}$ such that $\|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 < 1/|\mathcal{B}|$ and resort to an argument similar as the final calculations of the proof of Corollary 22:

$$\sum_{b \in \mathcal{B}} q_b \mathbf{1}_{b \leq r(\alpha) - 1/|\mathcal{B}|} < \frac{1}{|\mathcal{B}|} + \sum_{b \in \mathcal{B}} \frac{1}{|\mathcal{B}|} \mathbf{1}_{b \leq r(\alpha) - 1/|\mathcal{B}|} = \frac{1}{|\mathcal{B}|} + r(\alpha) - \frac{1}{|\mathcal{B}|} \leq \alpha.$$

Therefore, $\text{mscv}(\alpha, \mathbf{q}) \geq r(\alpha) - 1/|\mathcal{B}| \geq 0$, so that

$$\mathfrak{L}(C_{1-\text{mscv}(\alpha, \mathbf{q})}) \leq \mathfrak{L}(C_{1-r(\alpha)+1/|\mathcal{B}|}) = \mathfrak{L}(C_{1-\alpha+1/|\mathcal{B}|}).$$

The property above holds for all $\mathbf{q}$ such that $\|\mathbf{q} - \mathbf{q}_{\text{unif}}\|_1 < 1/|\mathcal{B}|$, which concludes the proof. $\blacksquare$

A.5 Proof of Theorem 34 of Section 6

The proof is obtained by adapting the proof provided for Theorem 19 in Appendix A.2, and is thus ultimately also an adaptation of the proof techniques by Bernstein et al. (2014).

Theorem 34 (Opportunistic ε -approachability for adaptive conformal inference)

Fix $\varepsilon \in (0, 1)$. The BO-ACI strategy of Algorithm 1, when tuned with $\alpha \in (0, 1)$ and run with an auxiliary ε -calibrated forecaster $\mathcal{F}$ relying on a disjoint ε -covering compatible with p^* in the sense of (17), is an ε -opportunistically approaching strategy in the following sense: for any opponent, for any convex subset $Q \subseteq \Delta(\mathcal{B})$,

$$\text{for all } t \geq 1, \quad d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.}$$

$$\text{entails} \quad \limsup_{T \rightarrow \infty} d(\bar{\mathbf{m}}_T, \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}) \leq (2L + 1)(\varepsilon + 2\sqrt{\varepsilon}) \quad \text{a.s.},$$

$$\text{where } \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}} = \text{cl}\left(\text{conv}\left(\{m^*(\mathbf{q}) : d(\mathbf{q}, Q) \leq \sqrt{\varepsilon}\}\right)\right).$$

Proof The proof follows the same structure as the one of Theorem 19, whose notation we continue to use (see Appendix A.2). In particular, we start with the triangular inequality

$$d(\bar{\mathbf{m}}_T, \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}) \leq \underbrace{\left\| \bar{\mathbf{m}}_T - \frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t) \right\|}_{\limsup \dots \leq (2L+1)\varepsilon} + \underbrace{d\left(\frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t), \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}\right)}_{\limsup \dots \leq 2(2L+1)\sqrt{\varepsilon}}; \quad (25)$$

then, each step term is almost surely bounded by the limit quantities indicated above.

Step 1. The following two rewritings only depended on the BO-ACI strategy of Algorithm 1, not on the auxiliary forecasting strategy, and they thus still hold:

$$\bar{\mathbf{m}}_T = \sum_{a \in \mathcal{A}} m\left(a, \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} \boldsymbol{\delta}_{b_t}\right) \quad \text{and} \quad \frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t) = \sum_{a \in \mathcal{A}} m\left(a, \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} \mathbf{z}_t\right),$$

where $\|m(a, b)\| \leq 2L + 1$ for all (a, b) , so that

$$\left\| \bar{\mathbf{m}}_T - \frac{1}{T} \sum_{t=1}^T m(a_t, \mathbf{z}_t) \right\| \leq (2L + 1) \sum_{a \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{\mathbf{z}_t \in R_a} (\mathbf{z}_t - \boldsymbol{\delta}_{b_t}) \right\|.$$

We now use the properties of the auxiliary forecasting strategy: the compatibility property (17) of the disjoint ε -covering $\mathcal{P} = \{P_k : k \in [N]\}$ implies that

$$\forall a \in \mathcal{A}, \quad R_a = \bigcup_{k : P_k \subseteq R_a} P_k,$$

so that, by triangular inequalities for the first inequality, by Definition 33 of ε -calibration for the equality and the final inequality,

$$\begin{aligned} \limsup_{T \rightarrow \infty} \sum_{a \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{z_t \in R_a} (z_t - \delta_{b_t}) \right\| &\leq \limsup_{T \rightarrow \infty} \sum_{k=1}^N \overbrace{\sum_{a \in \mathcal{A}} \mathbf{1}_{P_k \subseteq R_a}}^{=1} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{z_t \in P_k} (z_t - \delta_{b_t}) \right\| \\ &= \limsup_{T \rightarrow \infty} \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{c}_k - \delta_{b_t}) \right\| \leq \varepsilon \quad \text{a.s.} \end{aligned}$$

Collecting all elements, we proved the first limsup in (25).

Step 2. For the second terms in (25), we first note that by definition of action a_t played and by ε -calibration,

$$\forall t \geq 1, \quad m(a_t, z_t) = m^*(z_t) = m(a_t, \mathbf{c}_{k_t}).$$

We introduce the set of indices for which $\mathbf{c}_k$ belongs to the $\sqrt{\varepsilon}$ -neighborhood of Q ,

$$\mathcal{K}_\varepsilon = \{k \in [N] : d(\mathbf{c}_k, Q) \leq \sqrt{\varepsilon}\},$$

and show later that

$$\limsup_{T \rightarrow \infty} \frac{T - N_{\mathcal{K}_\varepsilon}(T)}{T} \leq \sqrt{\varepsilon} \quad \text{a.s.}, \quad \text{where} \quad N_{\mathcal{K}_\varepsilon}(T) = \sum_{k \in \mathcal{K}_\varepsilon} N_k(T). \quad (26)$$

Given that $\varepsilon \in (0, 1)$ by assumption, we have $\limsup N_{\mathcal{K}_\varepsilon}(T) \geq 1$, and for T such that $N_{\mathcal{K}_\varepsilon}(T) \geq 1$,

$$\begin{aligned} d\left(\frac{1}{T} \sum_{t=1}^T m(a_t, z_t), \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}\right) &= d\left(\frac{1}{T} \sum_{k \in [N]} N_k(T) m^*(\mathbf{c}_k), \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}\right) \\ &\leq \underbrace{\left\| \frac{1}{T} \sum_{k \in [N]} N_k(T) m^*(\mathbf{c}_k) - \frac{1}{N_{\mathcal{K}_\varepsilon}(T)} \sum_{k \in \mathcal{K}_\varepsilon} N_k(T) m^*(\mathbf{c}_k) \right\|}_{\limsup \dots \leq 2(2L+1)\sqrt{\varepsilon}} + \underbrace{d\left(\frac{1}{N_{\mathcal{K}_\varepsilon}(T)} \sum_{k \in \mathcal{K}_\varepsilon} N_k(T) m^*(\mathbf{c}_k), \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}}\right)}_{=0 \text{ by definition of } \mathcal{C}_{\alpha, Q}^{\sqrt{\varepsilon}} \text{ and } \mathcal{K}_\varepsilon}. \end{aligned}$$

The first term above is bounded as in the proof of Theorem 19, with $N_{\mathcal{K}_\varepsilon}(T)$ playing the role of $|\mathcal{V}_T(\eta)|$, by using the limit (26), which we prove next.

Introduce (as in the proof of Lemma 35) the notation, for each $k \in [N]$,

$$\bar{\mathbf{q}}_k(T) = \begin{cases} \frac{1}{N_k(T)} \sum_{t=1}^T \mathbf{1}_{k_t=k} \mathbf{q}_t, & \text{if } N_k(T) \stackrel{\text{def}}{=} \sum_{t=1}^T \mathbf{1}_{k_t=k} \geq 1, \\ \mathbf{0}, & \text{otherwise.} \end{cases}$$

For all k, T such that $N_k(T) \geq 1$, we have $d(\bar{\mathbf{q}}_k(T), Q) = 0$ almost surely. Indeed, the set Q , thus the function $d(\cdot, Q)$, are convex, and by assumption $d(\mathbf{q}_\tau, Q) = 0$ almost surely for all τ . Hence, by a triangular inequality, for all k, T such that $N_k(T) \geq 1$,

$$\sqrt{\varepsilon} \mathbf{1}_{d(\mathbf{c}_k, Q) > \sqrt{\varepsilon}} \leq d(\mathbf{c}_k, Q) \leq \|\mathbf{c}_k - \bar{\mathbf{q}}_k(T)\| + d(\bar{\mathbf{q}}_k(T), Q) = \|\mathbf{c}_k - \bar{\mathbf{q}}_k(T)\| \quad \text{a.s.}$$

We actually get for all pairs k, T , even when $N_k(T) = 0$, that

$$\sqrt{\varepsilon} N_k(T) \mathbf{1}_{d(\mathbf{c}_k, Q) > \sqrt{\varepsilon}} \leq N_k(T) \|\mathbf{c}_k - \bar{\mathbf{q}}_k(T)\| \quad \text{a.s.}$$

Therefore,

$$\begin{aligned} \sqrt{\varepsilon} \frac{T - N_{\mathcal{K}_\varepsilon}(T)}{T} &= \sqrt{\varepsilon} \sum_{k=1}^N \frac{N_k(T)}{T} \mathbf{1}_{d(\mathbf{c}_k, Q) > \sqrt{\varepsilon}} \\ &\leq \sum_{k=1}^N \frac{N_k(T)}{T} \|\mathbf{c}_k - \bar{\mathbf{q}}_k(T)\| = \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{c}_k - \mathbf{q}_t) \right\|. \end{aligned}$$

Now, by the ε -calibration property of Definition 33,

$$\limsup_{T \rightarrow \infty} \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{c}_k - \delta_{b_t}) \right\| \leq \varepsilon \quad \text{a.s.},$$

which by martingale convergence (e.g., repeated applications of the Hoeffding-Azuma inequality together with the Borel-Cantelli lemma) is equivalent to

$$\limsup_{T \rightarrow \infty} \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{k_t=k} (\mathbf{c}_k - \mathbf{q}_t) \right\| \leq \varepsilon \quad \text{a.s.}$$

Collecting all elements, we proved the limit (26), which concludes the proof. $\blacksquare$

Appendix B. Reminder of Classic Results in Online Adversarial Learning

This appendix restates classic results in online adversarial learning, i.e., in the settings where a learner faces an opponent reacting to the actions picked. More precisely,

- Appendix B.1 is devoted to Blackwell approachability in its original form (Blackwell, 1956);
- Appendix B.2 presents the extension by Bernstein et al. (2014) consisting of opportunistic approachability and which forms the key tool used in designing and analyzing the BO-ACI strategy of Algorithm 1.

We provide here mere statements of the results but Appendices C and Appendices D offer detailed proofs. The considered vector-valued payoff function will be denoted by $\mathbf{m}$, to distinguish it from the specific function m considered in the main body and in Appendix A.

B.1 Blackwell Approachability in Its Original Form

We recall Blackwell's approachability theory for general finite action sets $\mathcal{A}$ and $\mathcal{B}$ and for general vector-valued payoff functions $\mathbf{m} : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}^d$. In the main body, this theory is applied to the specific payoff function (6) but this appendix considers a general bounded vector-valued payoff function $\mathbf{m}$.

At each round $t \geq 1$, the learner and the opponent simultaneously pick actions $a_t \in \mathcal{A}$ and $b_t \in \mathcal{B}$. The learner obtains the payoff $\mathbf{m}(a_t, b_t)$ and observes it; unless explicitly stated (e.g., in the main body and in Appendix D), the learner does not have access to the action b_t of the opponent. The learner wants the average payoff to converge to a closed convex set $\mathcal{C} \subset \mathbb{R}^d$, referred to as the target set. Formally, the aim of the learner is to ensure that

$$\frac{1}{T} \sum_{t=1}^T \mathbf{m}(a_t, b_t) \stackrel{\text{def}}{=} \bar{\mathbf{m}}_T \longrightarrow \mathcal{C} \quad \text{a.s.} \quad \text{in the sense} \quad \lim_{T \rightarrow \infty} d(\bar{\mathbf{m}}_T, \mathcal{C}) = 0, \quad (27)$$

where, for $u \in \mathbb{R}^d$, the distance of u to the closed convex set $\mathcal{C}$ equals $d(u, \mathcal{C}) = \min_{c \in \mathcal{C}} \|u - c\|$.

The aim of the opponent is to prevent this convergence. Actually, the proof of Blackwell's theorem (Theorem 37 below) shows that either the learner can approach $\mathcal{C}$ or the opponent can approach the complement of a neighborhood of $\mathcal{C}$, i.e., make sure that

$$\liminf_{T \rightarrow \infty} d(\bar{\mathbf{m}}_T, \mathcal{C}) \geq \gamma > 0 \quad \text{a.s.},$$

for some $\gamma > 0$, irrespective of the strategy of the learner.

Blackwell's theorem provides a necessary and sufficient condition for the approachability of a closed convex set $\mathcal{C}$ in a finite game, based on the bilinear extension of $\mathbf{m}$: for all probability distributions $\mathbf{p} = (p_a)_{a \in \mathcal{A}} \in \Delta(\mathcal{A})$ and $\mathbf{q} = (q_b)_{b \in \mathcal{B}} \in \Delta(\mathcal{B})$,

$$\mathbf{m}(\mathbf{p}, \mathbf{q}) = \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} p_a q_b \mathbf{m}(a, b).$$

Theorem 37 (Approachability theorem, Blackwell, 1956) *A closed convex subset $\mathcal{C}$ of $\mathbb{R}^d$ is approachable by the learner, i.e., (27) may hold, if and only if*

$$\forall \mathbf{q} \in \Delta(\mathcal{B}), \exists \mathbf{p} \in \Delta(\mathcal{A}) \quad \text{s.t.} \quad \mathbf{m}(\mathbf{p}, \mathbf{q}) \in \mathcal{C}. \quad (28)$$

The condition (28) is referred to as Blackwell's dual condition. The proof of Theorem 37 is constructive: the explicit algorithm introduced therein is based on a geometric argument and is referred to as the primal Blackwell's approachability strategy. It consists, at round t , of drawing a_t independently at random according to a distribution $\mathbf{p}_t$ satisfying

$$\mathbf{p}_t \in \arg \min_{\mathbf{p} \in \Delta(\mathcal{A})} \max_{\mathbf{q} \in \Delta(\mathcal{B})} \langle \bar{\mathbf{m}}_{t-1} - \text{Proj}_{\mathcal{C}}(\bar{\mathbf{m}}_{t-1}), \mathbf{m}(\mathbf{p}, \mathbf{q}) \rangle, \quad (29)$$

where $\text{Proj}_{\mathcal{C}}$ is the projection in Euclidean norm onto $\mathcal{C}$.

Remark 38 *Theorem 37 mostly indicates whether a sequential-learning problem evaluated through average payoffs is feasible or not, but in general, the so-called primal approachability strategy (29) cannot be computed efficiently (often due to the projection onto $\mathcal{C}$) nor is the strategy achieving the optimal rates of convergence.*

Elements of proofs for Theorem 37. The sufficient part follows from arguments located in Appendix C: the strategy (29) is studied therein, for the case $Q = \Delta(\mathcal{B})$.

The necessity part works by contradiction: when there exists $\mathbf{q}^\dagger \in \Delta(\mathcal{B})$ such that $\mathbf{m}(\mathbf{p}, \mathbf{q}^\dagger) \notin \mathcal{C}$ for all $\mathbf{p} \in \Delta(\mathcal{A})$, then, since $\mathcal{C}$ is closed, there exists $\gamma > 0$ such that

$$\{\mathbf{m}(\mathbf{p}, \mathbf{q}^\dagger) : \mathbf{p} \in \Delta(\mathcal{A})\} \subseteq \overline{\mathcal{C}^\gamma},$$

where $\mathcal{C}^\gamma$ is the γ -neighborhood of $\mathcal{C}$ in Euclidean norm and $\overline{\mathcal{C}^\gamma}$ is the complement of $\mathcal{C}^\gamma$. Assume that the opponent draws the actions b_t i.i.d. according to $\mathbf{q}^\dagger$. For all strategies of the learner, by martingale convergence (e.g., repeated applications of Hoeffding's inequality together with the Borel-Cantelli lemma),

$$\lim_{T \rightarrow \infty} \left\| \mathbf{m}(\bar{\mathbf{p}}_T, \mathbf{q}^\dagger) - \frac{1}{T} \sum_{t=1}^T \mathbf{m}(a_t, b_t) \right\| = 0 \quad \text{a.s.},$$

where $\bar{\mathbf{p}}_T$ denotes the empirical distribution of the actions a_t played between rounds $t = 1$ and $t = T$. Thus,

$$\frac{1}{T} \sum_{t=1}^T \mathbf{m}(a_t, b_t) \longrightarrow \overline{\mathcal{C}^\gamma} \quad \text{a.s.}, \quad \text{as } T \rightarrow \infty.$$

In particular, the convergence to $\mathcal{C}$ stated in (27) cannot hold.

B.2 Opportunistic Approachability by Bernstein et al. (2014)

This section summarizes the setting and main result of Bernstein et al. (2014). The aim is to design a general strategy of the learner not only approaching a given approachable set $\mathcal{C} \subset \mathbb{R}^d$, but actually approaching a smaller target set $\mathcal{C}_Q \subseteq \mathcal{C}$ whenever the opponent is Q -restricted, with $Q \subseteq \Delta(\mathcal{B})$. We recall that the concept of Q -restricted opponent's play was introduced in Definition 12.

More formally, we consider the general framework of Section B.1 for a closed convex approachable set $\mathcal{C}$. The dual condition (28) in Blackwell's approachability theorem (see Theorem 37) ensures the existence of a response function $\mathbf{p}^* : \Delta(\mathcal{B}) \rightarrow \Delta(\mathcal{A})$ and of an associated goal function $\mathbf{m}^* : \Delta(\mathcal{B}) \rightarrow \mathcal{C}$ such that

$$\forall \mathbf{q} \in \Delta(\mathcal{B}), \quad \mathbf{m}^*(\mathbf{q}) \stackrel{\text{def}}{=} \mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) \in \mathcal{C}. \quad (30)$$

The response function $\mathbf{p}^*$ is not uniquely defined; the learner should pick a response function $\mathbf{p}^*$ such that the resulting payoff $\mathbf{m}^*(\mathbf{q})$ is desirable for any mixed action $\mathbf{q} \in \Delta(\mathcal{B})$ of the opponent.

We exploit this choice of $\mathbf{m}^*$ to define smaller target sets $\mathcal{C}_Q$ for Q -restricted opponents: namely, $\mathcal{C}_Q$ is based on the image of Q under the goal function $\mathbf{m}^*$, with some additional twists required for some technical reasons (taking closures⁴ of convex hulls and considering the non-increasing limit of images of η -neighborhoods of Q).

4. Bernstein et al. (2014) do not consider closures, which is not an issue in many cases but can be in some other pathological cases; we go over these issues in Appendix D.

Definition 39 (closed convex image under $\mathbf{m}^\star$) *The closed convex image of the convex subset $Q \subseteq \Delta(\mathcal{B})$ under the goal function $\mathbf{m}^\star$ is defined as*

$$\mathcal{C}_Q = \bigcap_{\eta > 0} \text{cl} \left(\text{conv} \left(\{ \mathbf{m}^\star(\mathbf{q}) : d(\mathbf{q}, Q) \leq \eta \} \right) \right).$$

We now state formally the property that strategies are expected to satisfy in this framework.

Definition 40 (opportunistic approachability) *A strategy is said to be opportunistically approaching a closed convex set $\mathcal{C}$ given a goal function $\mathbf{m}^\star$ if, for any convex subset $Q \subseteq \Delta(\mathcal{B})$, it approaches the closed convex image $\mathcal{C}_Q$ whenever the opponent is Q -restricted:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.} \quad \text{entails} \quad \lim_{T \rightarrow \infty} d(\bar{\mathbf{m}}_T, \mathcal{C}_Q) = 0 \quad \text{a.s.}$$

Such strategies exist, and are based on auxiliary calibrated forecasters (see Section 4.2 for a reminder on the latter concept). Bernstein et al. (2014) provide a general construction and study of opportunistically approaching strategies for general closed convex approachable sets $\mathcal{C}$, that comes with some necessary technicalities and is therefore rather long and complex. (They actually even cover the more general case of so-called D-sets $\mathcal{C}$.)

For the sake of completeness, we provide in Appendix D simplified proofs for this general construction.

Appendix C. On Q -Primal Strategies to Approach $\mathcal{C}_Q$ when Q is Known

This section goes over Remark 14 and does so both in some generality (with a general vector payoff function $\mathbf{m}$) and for the specific case of adaptive conformal inference (with the specific vector payoff function m introduced therein).

The notion of opportunistic approachability is designed to handle unknown convex subsets $Q \subseteq \Delta(\mathcal{B})$ and relies on a so-called dual strategy, exploiting a response function $\mathbf{p}^\star$, as introduced in Section B.2. If instead Q were known to the learner, one may want to adapt the primal approachability strategy (29) to directly approach $\mathcal{C}_Q$, and not only $\mathcal{C}$. In this section, we first define such a strategy, show (in Appendix C.2) that it indeed approaches $\mathcal{C}_Q$ whenever the opponent is Q -restricted, and characterize it in the context of adaptive conformal inference in the case of the exchangeable setting (in Appendix C.1).

General statement. Under the Blackwell condition (28), the Q -primal strategy is defined as follows: at round $t \geq 1$, draw a_t independently at random according to a distribution $\mathbf{p}_t$ satisfying

$$\mathbf{p}_t \in \arg \min_{\mathbf{p} \in \Delta(\mathcal{A})} \max_{\mathbf{q} \in \text{cl}(Q)} \langle \bar{\mathbf{m}}_{t-1} - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_{t-1}), \mathbf{m}(\mathbf{p}, \mathbf{q}) \rangle. \quad (31)$$

This strategy only differs from the original primal Blackwell's strategy (29) in that the maximum is taken over $\text{cl}(Q)$ instead of $\Delta(\mathcal{B})$ and in that the projection is taken onto $\mathcal{C}_Q$ instead of $\mathcal{C}$. Note that the closure $\text{cl}(Q)$ is used in (31), mostly for the sake of elegance of

the formula (to have a maximum instead of a mere supremum over Q), without changing the strategy in any way.

We state next that the Q -primal strategy approaches $\mathcal{C}_Q$ whenever the opponent is Q -restricted. This result generalizes the sufficient condition of Blackwell's approachability theorem (restated as Theorem 37); the latter indeed corresponds to the case $Q = \Delta(\mathcal{B})$.

Theorem 41 *If the opponent's play is Q -restricted, then the Q -primal strategy defined recursively by (31) approaches the target set $\mathcal{C}_Q$, i.e.,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.} \quad \text{entails} \quad \bar{\mathbf{m}}_T \longrightarrow \mathcal{C}_Q \quad \text{a.s.}$$

The theorem is proved below, in Appendix C.2. Its proof is an adaptation of the classic proof by Blackwell (1956) (see also the exposition by Mertens et al., 2015, Chapter 2), with some immediate adaptations to handle the Q -restriction on the opponent's play.

C.1 Specific statement in the exchangeable setting

We now instantiate the Q -primal strategy for adaptive conformal inference in the case of the exchangeable setting, where $Q = \{\mathbf{q}_{\text{unif}}\}$. We thus consider the specific vector payoff function m of Theorem 7.

When exchangeability is known to hold, the natural strategy is the classic conformal procedure, which plays $a_t = r(\alpha)$ at every round t . We show that the $\{\mathbf{q}_{\text{unif}}\}$ -primal strategy does not reduce to this procedure: see Example 6 below.

In contrast, the opportunistic approachability strategy of Algorithm 1 adapts to this favorable regime: whenever the calibrated forecaster outputs a forecast $\mathbf{z}_t$ sufficiently close to $\mathbf{q}_{\text{unif}}$ it picks $a_t = r(\alpha)$, exactly as in classic conformal prediction. Indeed, the final part of the proof of Corollary 22, located in Appendix A.4, shows that $\text{mscv}(\alpha, \mathbf{q}) = r(\alpha)$ for distributions $\mathbf{q}$ in a $\min\{\alpha - r(\alpha), r(\alpha) - \alpha + 1/|\mathcal{B}|\}$ -neighborhood of $\mathbf{q}_{\text{unif}}$ in ℓ_1 -norm.

Example 6 ($\{\mathbf{q}_{\text{unif}}\}$ -primal strategy) *Consider $\alpha \notin \mathcal{A}$. In the exchangeable setting, the $\{\mathbf{q}_{\text{unif}}\}$ -primal strategy may be written in closed form as follows (where quadrants are as in Figure 1):*

- If $\bar{\mathbf{m}}_{t-1} = (r(\alpha), \mathfrak{L}(C_{1-r(\alpha)}))^\top$, then $\mathbf{p}_t$ can be any distribution in $\Delta(\mathcal{A})$;
- Otherwise,

$$\mathbf{p}_t \begin{cases} = \delta_0, & \text{if } \bar{\mathbf{m}}_{t-1} \text{ is in the bottom right quadrant,} \\ = \delta_1, & \text{if } \bar{\mathbf{m}}_{t-1} \text{ is in the top left quadrant,} \\ \text{is any } \mathbf{p} \in \Delta(\mathcal{A}_t), & \text{if } \bar{\mathbf{m}}_{t-1} \text{ is in the bottom left or top right quadrant,} \end{cases}$$

where

$$\mathcal{A}_t = \arg \min_{a \in \mathcal{A}} \left\{ (\Pi_1(\bar{\mathbf{m}}_{t-1}) - r(\alpha))a + (\Pi_2(\bar{\mathbf{m}}_{t-1}) - \mathfrak{L}(C_{1-r(\alpha)})) \mathfrak{L}(C_{1-a}) \right\}$$

and $\Pi_1(\cdot)$ and $\Pi_2(\cdot)$ denote respectively the extraction of the first and second coordinates of a vector in $\mathbb{R}^2$.

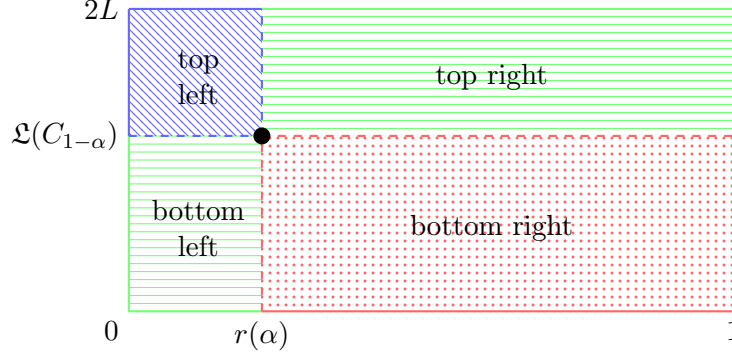


Figure 1: Decomposition of $[0, 1] \times [0, 2L]$ into quadrants and a central point; the blue and red dotted lines belong to their quadrants while the black dot is the singleton set $\mathcal{C}_{\alpha, Q}$ and does not belong to any of the four quadrants. The $\{\mathbf{q}_{\text{unif}}\}$ -primal strategy depends on the position of $\bar{\mathbf{m}}_{t-1}$ in this space.

Proof Fix $\alpha \notin \mathcal{A}$. In the exchangeable setting, we have $\mathcal{C}_{\alpha, Q} = \{r(\alpha)\} \times \{\mathfrak{L}(C_{1-r(\alpha)})\}$ since m^* is continuous at $\mathbf{q}_{\text{unif}}$ as $\alpha \notin \mathcal{A}$; see the proof of Corollary 22. Thus, we get that

$$\bar{\mathbf{m}}_{t-1} - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_{t-1}) = \begin{pmatrix} \Pi_1(\bar{\mathbf{m}}_{t-1}) - r(\alpha) \\ \Pi_2(\bar{\mathbf{m}}_{t-1}) - \mathfrak{L}(C_{1-r(\alpha)}) \end{pmatrix}.$$

Resorting to the definition of the payoff function together with the elementary fact that

$$\forall a \in \mathcal{A}, \quad \sum_{b \in \mathcal{B}} \mathbf{1}_{b \leq a} = (T_{\text{calib}} + 1) a,$$

we get $m(a, \mathbf{q}_{\text{unif}}) = (a, \mathfrak{L}(C_{1-a}))^\top$ for all $a \in \mathcal{A}$. Therefore, we obtain the following rewriting, for all $\mathbf{p} \in \Delta(\mathcal{A})$,

$$\begin{aligned} \langle \bar{\mathbf{m}}_{t-1} - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_{t-1}), m(\mathbf{p}, \mathbf{q}_{\text{unif}}) \rangle \\ = \sum_{a \in \mathcal{A}} p_a \left((\Pi_1(\bar{\mathbf{m}}_{t-1}) - r(\alpha))a + (\Pi_2(\bar{\mathbf{m}}_{t-1}) - \mathfrak{L}(C_{1-r(\alpha)})) \mathfrak{L}(C_{1-a}) \right). \end{aligned}$$

The claimed case decomposition comes from the fact that the above expression is linear in $\mathbf{p}$, so that $\mathbf{p}_t$ can be any distribution with support on the set $\mathcal{A}_t$ of minimizers of the following function:

$$a \in \mathcal{A} \mapsto (\Pi_1(\bar{\mathbf{m}}_{t-1}) - r(\alpha))a + (\Pi_2(\bar{\mathbf{m}}_{t-1}) - \mathfrak{L}(C_{1-r(\alpha)})) \mathfrak{L}(C_{1-a}).$$

In the top left and bottom right quadrants (blue and red regions in Figure 1), the above function is strictly monotone in a (we use here also that calibration scores are all different) and is thus minimized by either $a = 0$ or $a = 1$. For instance, if the first term is negative and the second term is positive (i.e., $\bar{\mathbf{m}}_{t-1}$ is in interior of the top left quadrant), then the

above function is decreasing in a and therefore minimized by $a = 1$; this argument extends to the boundary of that quadrant (when one of the terms is null, which corresponds to the dotted frontier lines in Figure 1). A symmetric argument holds for the bottom right quadrant and returns $a = 0$.

In the remaining cases, i.e., the green region in Figure 1, the above function is not necessarily monotone in a and may even have several minimizers, so that $\mathbf{p}_t$ can be any distribution supported on the set of minimizers. $\blacksquare$

C.2 Proof of Theorem 41

Theorem 41 *If the opponent's play is Q -restricted, then the Q -primal strategy defined recursively by (31) approaches the target set $\mathcal{C}_Q$, i.e.,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_t, Q) = 0 \quad \text{a.s.} \quad \text{entails} \quad \bar{\mathbf{m}}_T \longrightarrow \mathcal{C}_Q \quad \text{a.s.}$$

The proof is an adaptation of the classic proof by Blackwell (1956) (see also the exposition by Mertens et al., 2015, Chapter 2), with some immediate adaptations to handle the Q -restriction on the opponent's play.

Proof The proof is an immediate adaptation of the classic proof by Blackwell (1956) to handle the Q -restriction on the opponent's play.

We decompose the proof in four steps: we first use the dual condition together with a minimax argument to get a supporting-hyperplane inequality; second, we use this inequality together with an induction argument to derive an upper bound on the distance to the target set $\mathcal{C}_Q$; third, we invoke the Q -restriction property and get the convergence in probability (the adaptations are mainly located here); fourth, we conclude by studying the convergence of a non-negative supermartingale, (as in Mertens et al., 2015, Chapter 2), to get the desired almost-sure convergence.

Step 1. We show that, for all $t \geq 1$, the mixed action $\mathbf{p}_{t+1} \in \Delta(\mathcal{A})$ defined in (31) satisfies

$$\forall \mathbf{q} \in \text{cl}(Q), \quad \langle \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t), \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}) - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t) \rangle \leq 0. \quad (32)$$

Fix $t \geq 1$ and $\mathbf{q} \in \text{cl}(Q)$. The construction of the response function $\mathbf{p}^*$ in (30) and of the target set $\mathcal{C}_Q$ in Definition (39) ensure that $\mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) \in \mathcal{C}_Q$. Hence, by the characterization of the Euclidean projection onto a closed convex set (Hilbert projection theorem applied to the non-empty closed convex set $\mathcal{C}_Q$), we get that,

$$\langle \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t), \mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t) \rangle \leq 0.$$

In particular, it implies that the minimum over $\mathbf{p} \in \Delta(\mathcal{A})$ of the left-hand side of the above inequality is also non-positive. The latter holds for all $\mathbf{q} \in \text{cl}(Q)$, which by the Von Neumann minimax theorem, applied to the convex and compact sets $\Delta(\mathcal{A})$ and $\text{cl}(Q)$, entails that

$$\min_{\mathbf{p} \in \Delta(\mathcal{A})} \max_{\mathbf{q} \in \text{cl}(Q)} \langle \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t), \mathbf{m}(\mathbf{p}, \mathbf{q}) - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t) \rangle \leq 0.$$

The definition of $\mathbf{p}_{t+1}$ in (31) concludes the first step of the proof.

Step 2. We denote $d_t = d(\bar{\mathbf{m}}_t, \mathcal{C}_Q)$, for all $t \geq 1$, and show that, for all $T \geq 1$,

$$\mathbb{E}[d_{T+1}^2] \leq \frac{2}{T+1} \|\mathbf{m}\|_\infty^2 + \frac{2}{T(T+1)} \sum_{t=1}^T t \mathbb{E}[\nu_t(Q)]. \quad (33)$$

where $\|\mathbf{m}\|_\infty = \max\{\|\mathbf{m}(a, b)\| : (a, b) \in \mathcal{A} \times \mathcal{B}\}$ is the uniform bound on the payoff function and $\nu_t(Q) = \langle \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}_{t+1}) - \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}'_{t+1}), \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t) \rangle$, with $\mathbf{q}'_{t+1}$ being the orthogonal projection of $\mathbf{q}_{t+1} \in \Delta(\mathcal{B})$ onto $\text{cl}(Q)$.

Fix $t \geq 1$. We first use that $d_{t+1} \leq \|\bar{\mathbf{m}}_{t+1} - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t)\|$, then that $\|\bar{\mathbf{m}}_{t+1} - \bar{\mathbf{m}}_t\| = \frac{1}{t+1} \|\mathbf{m}(a_{t+1}, b_{t+1}) - \bar{\mathbf{m}}_t\| \leq \frac{2}{t+1} \|\mathbf{m}\|_\infty$, and get the following recurrent inequality:

$$d_{t+1}^2 \leq d_t^2 + \frac{2}{t+1} \|\mathbf{m}\|_\infty^2 + \frac{2}{t+1} \langle \mathbf{m}(a_{t+1}, b_{t+1}) - \bar{\mathbf{m}}_t, \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t) \rangle. \quad (34)$$

Let $c_t = \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t)$. By a simple algebraic manipulation, we get that

$$\begin{aligned} \langle \mathbf{m}(a_{t+1}, b_{t+1}) - \bar{\mathbf{m}}_t, c_t \rangle &= \langle \mathbf{m}(a_{t+1}, b_{t+1}) - \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}_{t+1}), c_t \rangle \\ &\quad + \underbrace{\langle \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}_{t+1}) - \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}'_{t+1}), c_t \rangle}_{\leq 0 \text{ by inequality (32), obtained in Step 1}} + \langle \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}'_{t+1}) - \bar{\mathbf{m}}_t, c_t \rangle - d_t^2. \end{aligned}$$

Let $\mathcal{F}_t = \sigma((a_\tau, b_\tau)_{\tau \leq t})$, be the natural filtration so that $\mathbf{p}_{t+1}$, d_t and $\bar{\mathbf{m}}_t$ are $\mathcal{F}_t$ -measurable. Since $\mathbb{E}[\langle \mathbf{m}(a_{t+1}, b_{t+1}) - \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}_{t+1}), c_t \rangle | \mathcal{F}_t] = 0$ in the above decomposition, we consider the conditional expectation with respect to $\mathcal{F}_t$ in (34), and get that

$$\mathbb{E}[d_{t+1}^2 | \mathcal{F}_t] \leq \left(1 - \frac{2}{t+1}\right) d_t^2 + \frac{2}{(t+1)^2} \|\mathbf{m}\|_\infty^2 + \frac{2}{t+1} \mathbb{E}[\nu_t(Q) | \mathcal{F}_t]. \quad (35)$$

We can now unfold this induction inequality to get to the desired bound (33). Let $u_t = \mathbb{E}[d_t^2]$ and $v_t = \frac{2}{(t+1)^2} \|\mathbf{m}\|_\infty^2 + \frac{2}{t+1} \mathbb{E}[\nu_t(Q)]$. We take the expectation in (35), and get that

$$u_{t+1} \leq \left(1 - \frac{2}{t+1}\right) u_t + v_t = \frac{t-1}{t+1} u_t + v_t. \quad (36)$$

By induction, we show that the following inequality holds, for all $T \geq 1$:

$$u_{T+1} \leq \sum_{t=1}^T \frac{t(t+1)}{T(T+1)} v_t. \quad (37)$$

Indeed, (37) holds for $T = 1$ and assume it holds at round $T \geq 1$, then by (36)

$$u_{T+2} \leq \frac{T}{T+2} \sum_{t=1}^T v_t \frac{t(t+1)}{T(T+1)} + v_{T+1} \leq \sum_{t=1}^{T+1} v_t \frac{t(t+1)}{(T+1)(T+2)},$$

which concludes the proof by induction and also directly implies the desired bound (33).

Step 3. We invoke the Q -restriction property on the opponent's play, and show that

$$\lim_{T \rightarrow \infty} \sum_{t=1}^T \frac{2}{t+1} \nu_t(Q) = 0 \quad \text{a.s.}, \quad \text{and thus} \quad \lim_{T \rightarrow \infty} \mathbb{E}[d_T^2] = 0 \quad \text{a.s.}$$

By standard norm inequalities, we have that, for all $\mathbf{q}, \mathbf{q}' \in \Delta(\mathcal{B})$, and all $t \geq 1$,

$$\|\mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}) - \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}')\| \leq \|\mathbf{m}\|_\infty \|\mathbf{q} - \mathbf{q}'\|_1 \leq \|\mathbf{m}\|_\infty \sqrt{|\mathcal{B}|} d(\mathbf{q}, \mathbf{q}').$$

Therefore, by Cauchy-Schwarz inequality, we obtain that

$$\begin{aligned} \nu_t(Q) &= \langle \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}_{t+1}) - \mathbf{m}(\mathbf{p}_{t+1}, \mathbf{q}'_{t+1}), \bar{\mathbf{m}}_t - \text{Proj}_{\mathcal{C}_Q}(\bar{\mathbf{m}}_t) \rangle \\ &\leq (2\|\mathbf{m}\|_\infty^2 \sqrt{|\mathcal{B}|}) d(\mathbf{q}_{t+1}, Q) \end{aligned}$$

As a consequence, we get that, for all $T \geq 1$,

$$\sum_{t=1}^T \frac{2}{t+1} \nu_t(Q) \leq (4\|\mathbf{m}\|_\infty^2 \sqrt{|\mathcal{B}|}) \frac{1}{T} \sum_{t=1}^T d(\mathbf{q}_{t+1}, Q),$$

which converges to 0 almost surely by the Q -restriction property on the opponent's play. Thus, by the upper bound (33) obtained in Step 2, we also get that d_T^2 converges to 0 in probability, i.e., $\lim_{T \rightarrow \infty} \mathbb{E}[d_T^2] \rightarrow 0$.

Step 4. Finally, we show that d_t converges to 0 almost surely. To do so, we consider the following non-negative supermartingale $(w_t)_{t \geq 1}$, that uniformly bounds d_t^2 .

$$w_t = d_t^2 + \sum_{\tau=t}^{\infty} \frac{2}{(\tau+1)^2} \|\mathbf{m}\|_\infty^2 + \sum_{\tau=t}^{\infty} \frac{2}{\tau+1} \mathbb{E}[\nu_\tau(Q) | \mathcal{F}_t]. \quad (38)$$

We first check that $(w_t)_{t \geq 1}$ is indeed a non-negative supermartingale: we take the conditional expectation in (38), and use the tower rule $\mathbb{E}[\mathbb{E}[\cdots | \mathcal{F}_{t+1}] | \mathcal{F}_t] = \mathbb{E}[\cdots | \mathcal{F}_t]$, together with the recurrent inequality (35) obtained in Step 2 to get that

$$\begin{aligned} \mathbb{E}[w_{t+1} | \mathcal{F}_t] &= \mathbb{E}[d_{t+1}^2 | \mathcal{F}_t] + \sum_{\tau=t+1}^{\infty} \frac{2}{(\tau+1)^2} \|\mathbf{m}\|_\infty^2 + \sum_{\tau=t+1}^{\infty} \frac{2}{\tau+1} \mathbb{E}[\nu_\tau(Q) | \mathcal{F}_t] \\ &\leq d_t^2 + \sum_{\tau=t}^{\infty} \frac{2}{(\tau+1)^2} \|\mathbf{m}\|_\infty^2 + \sum_{\tau=t}^{\infty} \frac{2}{\tau+1} \mathbb{E}[\nu_\tau(Q) | \mathcal{F}_t] = w_t. \end{aligned}$$

The supermartingale is defined for all $t \geq 1$ as the sum of d_t^2 , which converges to 0 in probability by Step 3, and two remainder terms of convergent series (the convergence of the second series is also granted by Step 3), so that w_t converges to 0 in probability.

Finally, by the martingale convergence theorem, w_t converges almost surely to a finite limit, that is necessarily the same as the limit in probability, i.e., 0, which concludes the proof since $d_t^2 \leq w_t$, for all $t \geq 1$. $\blacksquare$

Appendix D. A Discussion and Simpler Proofs for the Opportunistic Approachability Theory by Bernstein et al. (2014)

We consider an extension of the general setting of Appendix B.2 that is designed to handle possibly non-convex target sets (known as D-sets), just as in Bernstein et al. (2014). The point of this appendix is to reprove the main result of the latter article, which corresponds to Theorem 42 of Appendix D.1 (and is actually a general version of the specific convergence result established in Theorem 19). Theorem 42 provides some general construction and analysis of opportunistically approaching strategies in the sense of Definition 40 (extended to D-sets).

There are two differences between the original version of the result by Bernstein et al. (2014) and our version in Theorem 42.

First, we require that closures are taken in the defining expression of $\mathcal{C}_Q$ (see Definition 39, unlike Bernstein et al., 2014, Definition 8), i.e., we shoot for a slightly less ambitious target. However, we prove in Appendix D.3 the necessity of taking these closures, through a counter-example, while also explaining in Appendix D.2 that taking or not taking closures is indifferent in several natural cases.

Second, we relax the assumption on the response function $\mathbf{p}^*$ into a mere measurability requirement (Bernstein et al., 2014, Theorem 4 assumed that $\mathbf{p}^*$ is piecewise continuous).

Setting, with D-sets. The formal setting throughout this appendix is the following. Let $\mathcal{A}$ and $\mathcal{B}$ be finite action sets, $\mathbf{m} : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}^d$ be a general vector-valued payoff function, and $\mathcal{C} \subset \mathbb{R}^d$ be a general D-set.

A closed set $\mathcal{C}$ is called a D-set if for all $\mathbf{q} \in \Delta(\mathcal{B})$, there exists $\mathbf{p} \in \Delta(\mathcal{A})$ such that $\mathbf{m}(\mathbf{p}, \mathbf{q}) \in \mathcal{C}$. For any D-set $\mathcal{C}$, we can thus define a response function $\mathbf{p}^* : \Delta(\mathcal{B}) \rightarrow \Delta(\mathcal{A})$ and an associated goal function $\mathbf{m}^* : \Delta(\mathcal{B}) \rightarrow \mathcal{C}$ such that

$$\forall \mathbf{q} \in \Delta(\mathcal{B}), \quad \mathbf{m}^*(\mathbf{q}) \stackrel{\text{def}}{=} \mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) \in \mathcal{C}. \quad (39)$$

This construction generalizes (30) to non-convex target sets. The response function $\mathbf{p}^*$ defined in Equation (39) will only be required to be measurable.

A repeated game takes place. At each round $t \geq 1$, the learner and the opponent simultaneously pick actions $a_t \in \mathcal{A}$ and $b_t \in \mathcal{B}$. The learner obtains the payoff $\mathbf{m}(a_t, b_t)$ and observes b_t , not only $\mathbf{m}(a_t, b_t)$ as in Section B.1. (This is because the strategy considered in Theorem 42 relies on an auxiliary calibrated strategy $\mathcal{F}$, forecasting the b_t .)

D.1 General Theory, with Closures

For the sake of clarity, we recast the general statement of the opportunistic approachability procedure in Algorithm 2, as introduced by Bernstein et al. (2014).

Theorem 42 (General opportunistic approachability) *If the response function $\mathbf{p}^* : \Delta(\mathcal{B}) \rightarrow \Delta(\mathcal{A})$ satisfying the dual condition (39) is measurable, then the strategy stated in Algorithm 2, when run with a calibrated forecaster, is an opportunistically approaching strategy for the D-set $\mathcal{C}$, in the sense of Definition 40, namely: for any convex subset $Q \subseteq \Delta(\mathcal{B})$, whenever the opponent is Q -restricted,*

$$\bar{\mathbf{m}}_T \longrightarrow \mathcal{C}_Q \quad \text{a.s.,} \quad \text{as } T \rightarrow \infty.$$

Algorithm 2 General opportunistic approachability procedure

Inputs: response function $\mathbf{p}^*$; auxiliary sequential forecaster $\mathcal{F}$ with outputs in $\Delta(\mathcal{B})$

Initialization: play a_1 uniformly at random over $\mathcal{A}$

At time steps $t = 2, 3, \dots$:

1. Get from $\mathcal{F}$ a probabilistic forecast $\mathbf{z}_t \in \Delta(\mathcal{B})$ of the opponent's pure action
 2. Play $a_t \sim \mathbf{p}^*(\mathbf{z}_t)$
 3. Receive $\mathbf{m}(a_t, b_t)$, observe $\mathbf{m}(a_t, b_t)$ and b_t , and feed $\mathcal{F}$ with b_t
-

Proof The proof follows the same lines and uses the same notation as the proof of Theorem 19 in Appendix A.2, the only change being the choice of the partition in the first step. In particular, we start with a similar triangular inequality,

$$d(\bar{\mathbf{m}}_T, \mathcal{C}_Q) \leq \left\| \bar{\mathbf{m}}_T - \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t) \right\| + d\left(\frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t), \mathcal{C}_Q\right), \quad (40)$$

where the convergence of the second term, involving the quantities $\mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t) = \mathbf{m}^*(\mathbf{z}_t)$, is ensured by Equation (23) and Appendix A.3, as both relied on general arguments (not involving the specific definition of $\mathbf{m}$ in Appendix A.2).

Therefore, we only need to show the convergence of the first term in Equation (40). First, by martingale convergence (e.g., the Hoeffding-Azuma inequality together with the Borel-Cantelli lemma),

$$\limsup_{T \rightarrow \infty} \left\| \bar{\mathbf{m}}_T - \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t) \right\| = \limsup_{T \rightarrow \infty} \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), b_t) - \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t) \right\|.$$

Fix $\varepsilon > 0$, and consider a disjoint ε -covering $\{P_k : k \in [N]\}$ of $\Delta(\mathcal{A})$ as per Definition 32, with associated centers $\{\mathbf{c}_k : k \in [N]\}$. This is the main change with respect to the proofs of Theorems 19 and 34: the covering is now on $\Delta(\mathcal{A})$, and not on $\Delta(\mathcal{B})$. For each $k \in [N]$, we define the preimage of P_k by $\mathbf{p}^*$ as $Q_k = \{\mathbf{q} \in \Delta(\mathcal{B}) : \mathbf{p}^*(\mathbf{q}) \in P_k\}$; by measurability of $\mathbf{p}^*$, the subset Q_k is also measurable. We group rounds according to the disjoint ε -covering:

$$\frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t) = \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^N \mathbf{1}_{\mathbf{z}_t \in Q_k} \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t).$$

We recall that for all $k \in [N]$,

$$\forall \mathbf{q} \in Q_k, \quad \|\mathbf{p}^*(\mathbf{q}) - \mathbf{c}_k\|_1 \leq \varepsilon, \quad \text{which entails} \quad \|\mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) - \mathbf{m}(\mathbf{c}_k, \mathbf{q})\| \leq \rho\varepsilon,$$

where we denoted $\rho = \max\{\|\mathbf{m}(a, b)\| : (a, b) \in \mathcal{A} \times \mathcal{B}\} < \infty$. Therefore, by a triangular inequality,

$$\left\| \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^N \mathbf{1}_{\mathbf{z}_t \in Q_k} \mathbf{m}(\mathbf{p}^*(\mathbf{z}_t), \mathbf{z}_t) - \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^N \mathbf{1}_{\mathbf{z}_t \in Q_k} \mathbf{m}(\mathbf{c}_k, \mathbf{z}_t) \right\| \leq \rho\varepsilon.$$

A similar argument may be applied to the average of the $\mathbf{m}(\mathbf{p}^*(z_t), b_t)$, and we get, by a triangular inequality, that the quantity of interest is bounded as

$$\begin{aligned} & \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(z_t), b_t) - \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(z_t), z_t) \right\| \\ & \leq 2\rho\varepsilon + \left\| \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^N \mathbf{1}_{z_t \in Q_k} \mathbf{m}(c_k, b_t) - \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^N \mathbf{1}_{z_t \in Q_k} \mathbf{m}(c_k, z_t) \right\| \\ & \leq 2\rho\varepsilon + \rho \sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{z_t \in Q_k} (\delta_{b_t} - z_t) \right\|_1, \end{aligned}$$

where the second inequality follows again by a triangular inequality and by the boundedness of $\mathbf{m}$. Now, by a triangular inequality, by a standard norm inequality, and by N applications of the definition of the calibration of $\mathcal{F}$ (this is where we use that the Q_k are measurable),

$$\sum_{k=1}^N \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{1}_{z_t \in Q_k} (\delta_{b_t} - z_t) \right\|_1 \leq \sqrt{|\mathcal{B}|} \sum_{k=1}^N \left\| \sum_{t=1}^T \mathbf{1}_{z_t \in Q_k} (z_t - \delta_{b_t}) \right\| \longrightarrow 0 \quad \text{a.s.}$$

Collecting all elements, we showed so far that for all $\varepsilon > 0$,

$$\limsup_{T \rightarrow \infty} \left\| \bar{\mathbf{m}}_T - \frac{1}{T} \sum_{t=1}^T \mathbf{m}(\mathbf{p}^*(z_t), z_t) \right\| \leq 2\rho\varepsilon \quad \text{a.s.},$$

from which the desired conclusion follows, by letting $\varepsilon \rightarrow 0$. ■

D.2 Special Case where Closures Are Not Necessary

Theorem 42 and the underlying Definitions 39 and 40 entail the convergence of $\bar{\mathbf{m}}_T$ to

$$\mathcal{C}_Q = \bigcap_{\eta > 0} \text{cl} \left(\text{conv} \left(\{ \mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in \Delta(\mathcal{B}) \text{ s.t. } d(\mathbf{q}, Q) \leq \eta \} \right) \right),$$

where $\mathbf{m}^*(\mathbf{q}) = \mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q})$ and when the opponent plays in a Q -restricted way.

Sometimes (but not always, see Appendix D.3), convergence to a smaller target set $\mathcal{C}'_Q \subseteq \mathcal{C}_Q$ may be achieved; this set $\mathcal{C}'_Q$ is defined by not taking closures in the definition of $\mathcal{C}_Q$ and is the set considered in Bernstein et al. (2014). Formally,

$$\mathcal{C}'_Q = \bigcap_{\eta > 0} \text{conv} \left(\{ \mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in \Delta(\mathcal{B}) \text{ s.t. } d(\mathbf{q}, Q) \leq \eta \} \right). \quad (41)$$

We may indifferently consider $\mathcal{C}'_Q$ or $\text{cl}(\mathcal{C}'_Q)$, as the distance of a point to a set is equal to the distance of a point to the closure of that set. The closure $\text{cl}(\mathcal{C}'_Q)$ of the intersection of some sets is always contained in the intersection $\mathcal{C}_Q$ of the closures of these sets. The question is to determine when the converse is true. We recall that the dimension of a convex set C is the dimension of the affine subspace that it generates.

Lemma 43 *If $\text{int}(C'_Q) \neq \emptyset$, then $\text{cl}(C'_Q) = C_Q$, and thus, the convergence of $\overline{\mathbf{m}}_T$ to C_Q in Theorem 42 corresponds exactly to the convergence of $\overline{\mathbf{m}}_T$ to C'_Q .*

The equality holds more generally if the dimension of C'_Q equals the limit of the dimensions of the sets C_n defined in the proof.

Proof By nestedness, it suffices to apply the general result of Lemma 44 (which we state and prove next) to the sequences of convex sets

$$C_n = \text{conv} \left(\left\{ \mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in \Delta(\mathcal{B}) \text{ s.t. } d(\mathbf{q}, Q) \leq 1/n \right\} \right),$$

indexed by $n \geq 1$. ■

The following result is perhaps a well-known result but we could find no exactly similar statement of it; therefore, we provide a (simple but) complete proof thereof.

Lemma 44 *Let $(C_n)_{n \geq 1}$ be a sequence of convex subsets of $\mathbb{R}^d$ such that its intersection C has a non-empty interior; then the closure of C is the intersection of the closures of the C_n , i.e.,*

$$\text{let } C = \bigcap_{n \geq 1} C_n : \quad \text{int}(C) \neq \emptyset \quad \implies \quad \text{cl}(C) = \bigcap_{n \geq 1} \text{cl}(C_n).$$

The result holds more generally if the convex sets C_n are nested and if the limit of their dimensions as $n \rightarrow \infty$ equals the dimension of C .

Proof We prove the first statement of the lemma, under the condition that $\text{int}(C) \neq \emptyset$. As the intersection of closed sets always contains the closure of the intersection of sets (with no additional condition), it suffices to prove that

$$\text{cl}(C) \supseteq \bigcap_{n \geq 1} \text{cl}(C_n).$$

We do so by contradiction: suppose that there exists $x \in \bigcap_{n \geq 1} \text{cl}(C_n)$ such that $x \notin \text{cl}(C)$.

By the strong separation theorem for convex sets (e.g., Rockafellar and Wets, 1998, Theorem 2.39, also called the Hahn-Banach theorem), applied to the closed convex set $\text{cl}(C)$ and to the singleton $\{x\}$, which is a compact convex set, there exists a hyperplane $\mathcal{H} = \{u \in \mathbb{R}^d : h(u) = 0\}$ parameterized by an affine function h that strictly separates x from $\text{cl}(C)$, i.e., there exists $\delta > 0$ such that

$$\{x\} \subset \mathcal{H}^+ \stackrel{\text{def}}{=} \{u \in \mathbb{R}^d : h(u) > \delta\} \quad \text{and} \quad \text{cl}(C) \subseteq \mathcal{H}^- \stackrel{\text{def}}{=} \{u \in \mathbb{R}^d : h(u) < -\delta\}.$$

The idea of the proof consists in constructing a point z that belongs to $\mathcal{H}$ and to $\mathcal{H}^-$, which is impossible.

We use first the assumption of non-empty interior: take $y \in \text{int}(C)$. By definition, $C \subseteq C_n$ for all $n \geq 1$, and we thus have $y \in \text{int}(C_n)$ for all $n \geq 1$. Moreover, by design, $x \in \text{cl}(C_n)$ for all $n \geq 1$, and therefore $[y, x] \subseteq \text{int}(C_n)$ by convexity of C_n (this is a classic

property, see Lemma 45 below). Second, $y \in C$ thus $y \in \mathcal{H}^-$ while $x \in \mathcal{H}^+$, so that the segment $[y, x]$ intersects in particular the hyperplane $\mathcal{H}$: there exists $z \in (y, x) \cap \mathcal{H}$.

Finally, we use the inclusion $[y, x] \subseteq \text{int}(C_n)$ for all $n \geq 1$ to get that $z \in (y, x)$ also belongs to $\text{int}(C_n)$ thus to C_n , for all $n \geq 1$, and in turn, z belongs to C . This creates the contradiction: indeed, z belongs to $\mathcal{H}$ and was proved to belong to $C \subseteq \mathcal{H}^-$.

The second statement of the lemma relies on the following facts. Denote by $\mathcal{H}$ the (affine) subspace generated by C ; it can be identified with some $\mathbb{R}^{d'}$ space, up to some change of coordinates. By assumption, the convex sets C_n are nested, thus the sequence of their dimensions is a non-increasing, integer-valued sequence. That its limit equals the dimension of C means that there exists $N \geq 1$ such that C and C_n are subsets of $\mathcal{H}$ for all $n \geq N$. Indeed, C is included in C_n , and thus the affine space generated by C is included in the affine space generated by C_n ; as these two spaces have the same dimension, they are equal. Now, C has a non-empty interior when seen as a subset of $\mathcal{H}$: this follows from the general fact that a non-empty convex set has always a non-empty interior in the affine subspace it generates (e.g., Rockafellar, 1997, Theorem 6.2). Therefore, the second statement of the lemma follows from the first one, applied to the sequence $(C_{N+n})_{n \geq 1}$ of convex subsets of some $\mathbb{R}^{d'}$ space. $\blacksquare$

Lemma 45 (see, e.g., Rockafellar, 1997, Theorem 6.1) *Consider a convex $C \subseteq \mathbb{R}^d$ with non-empty interior; then,*

$$\forall x \in \text{cl}(C), \quad \forall y \in \text{int}(C), \quad [y, x] \subseteq C.$$

Proof Fix $w \in [y, x]$: there exists $\lambda \in [0, 1]$ such that $w = (1 - \lambda)y + \lambda x$. Since $y \in \text{int}(C)$, there exists $\varepsilon > 0$ such that $B(y, \varepsilon) \subseteq C$; in addition, the linear mapping $u \mapsto (1 - \lambda)u + \lambda x$ is a homothety of center x and ratio $1 - \lambda$ that maps the ball $B(y, \varepsilon)$ to the ball $B(w, (1 - \lambda)\varepsilon)$, which is included in $\text{cl}(C)$ by convexity of the latter. Therefore, the ball $B(w, (1 - \lambda)\varepsilon/2)$ is included in C , and thus $w \in \text{int}(C)$, as announced. $\blacksquare$

D.3 Counter-example for the General Necessity of Closures

In general, we do not have that $\text{cl}(\mathcal{C}'_Q)$ equals $\mathcal{C}_Q$. Bernstein et al. (2014, proof of Corollary 2) directly claim (with no proof and under the additional assumption that $\mathbf{p}^*$ is piecewise continuous) that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) = 0 \quad \text{entails} \quad d\left(\frac{1}{T} \sum_{t=1}^T \mathbf{m}^*(\mathbf{z}_t), \mathcal{C}'_Q\right) \rightarrow 0, \quad (42)$$

where $(\mathbf{z}_t)_{t \geq 1}$ is a sequence in $\Delta(\mathcal{B})$ and where we recall that $\mathcal{C}'_Q$ was defined in (41).

However, we found a counter-example showing that the above implication does not hold in general. Given the results of Appendix D.2, such a counter-example requires a gap in the limit for the dimensions of the convex sets involved: the limit of the dimensions of the

$$\mathcal{M}_Q^\eta = \text{conv}\left(\{\mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in \Delta(\mathcal{B}) \text{ s.t. } d(\mathbf{q}, Q) \leq \eta\}\right) \quad (43)$$

must be strictly larger than the dimension of their intersection $\mathcal{C}'_Q = \bigcap_{\eta>0} \mathcal{M}_Q^\eta$.

We have $\mathcal{C}_Q^\eta = \text{cl}(\mathcal{M}_Q^\eta)$ with the notation of Appendix A.3 but we omit the closures in this section. We build a simple counter-example in $\mathbb{R}^2$, where the $\mathcal{M}_Q^\eta$ are of dimension 2 (i.e., span the entire space) while $\mathcal{C}'_Q$ is of dimension 1 (i.e., it generates a line).

Remark 46 *The counter-example was inspired by the reward function defined for ACI in Section 2; however, for the sake of simplicity, we present it with explicit scalar values.*

Setting. We consider the action sets $\mathcal{A} = \{1, 2, 3\}$ and $\mathcal{B} = \{1, 2, 3\}$, and the following reward function $\mathbf{m} : \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}^2$, which we describe by listing all its values in a 3×3 matrix, where rows correspond to the learner playing in $\mathcal{A}$ and columns to the opponent picking actions in $\mathcal{B}$:

$$\begin{bmatrix} \mathbf{m}(1, 1) & \mathbf{m}(1, 2) & \mathbf{m}(1, 3) \\ \mathbf{m}(2, 1) & \mathbf{m}(2, 2) & \mathbf{m}(2, 3) \\ \mathbf{m}(3, 1) & \mathbf{m}(3, 2) & \mathbf{m}(3, 3) \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} & \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} & \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \\ \begin{pmatrix} -1 \\ 3 \end{pmatrix} & \begin{pmatrix} 0 \\ 3 \end{pmatrix} & \begin{pmatrix} 0 \\ 3 \end{pmatrix} \\ \begin{pmatrix} 1 \\ 1 \end{pmatrix} & \begin{pmatrix} 1 \\ 1 \end{pmatrix} & \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{bmatrix}. \quad (44)$$

The objective is to approach $\mathcal{C} = [-1, 0.2] \times [1, 3] \subseteq \mathbb{R}^2$.

Response function. We describe a (valid but not unique) response function $\mathbf{p}^*$ for the row player and its associated payoff function $\mathbf{m}^*$ defined by $\mathbf{q} \in \Delta(\mathcal{B}) \mapsto \mathbf{m}^*(\mathbf{q}) = \mathbf{m}(\mathbf{p}^*(\mathbf{q}), \mathbf{q}) \in \mathcal{C}$; namely, for all $\mathbf{q} \in \Delta(\mathcal{B})$,

$$\begin{cases} \mathbf{p}^*(\mathbf{q}) = \delta_1 & \text{and} & \mathbf{m}^*(\mathbf{q}) = \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \in \mathcal{C} & \text{if } q_1 = 0 \text{ and } q_1 + q_2 > 0.2, \\ \mathbf{p}^*(\mathbf{q}) = \delta_2 & \text{and} & \mathbf{m}^*(\mathbf{q}) = \begin{pmatrix} -q_1 \\ 3 \end{pmatrix} \in \mathcal{C} & \text{if } q_1 > 0 \text{ and } q_1 + q_2 > 0.2, \\ \mathbf{p}^*(\mathbf{q}) = \delta_3 & \text{and} & \mathbf{m}^*(\mathbf{q}) = \begin{pmatrix} q_1 + q_2 \\ 1 \end{pmatrix} \in \mathcal{C} & \text{if } q_1 + q_2 \leq 0.2. \end{cases}$$

Note that $\mathbf{p}^*$ is piecewise constant, and thus both measurable and piecewise continuous, as required by Theorem 42 and Bernstein et al. (2014, Corollary 2), respectively.

Set of Q -restrictions. Finally, we introduce the convex subset Q of $\Delta(\mathcal{B})$ defined by

$$Q = \{\mathbf{q} \in \Delta(\mathcal{B}) : q_1 = 0, q_2 \geq 0.2\}.$$

The core property of Q is that its boundary contains the discontinuity point $\mathbf{q} = (0, 0.2, 0.8)^\top$ of the response function $\mathbf{p}^*$, and that any neighborhood of $(0, 0.2, 0.8)^\top$ contains distributions $\mathbf{q}$ leading to any of the three possible outcomes of $\mathbf{p}^*$.

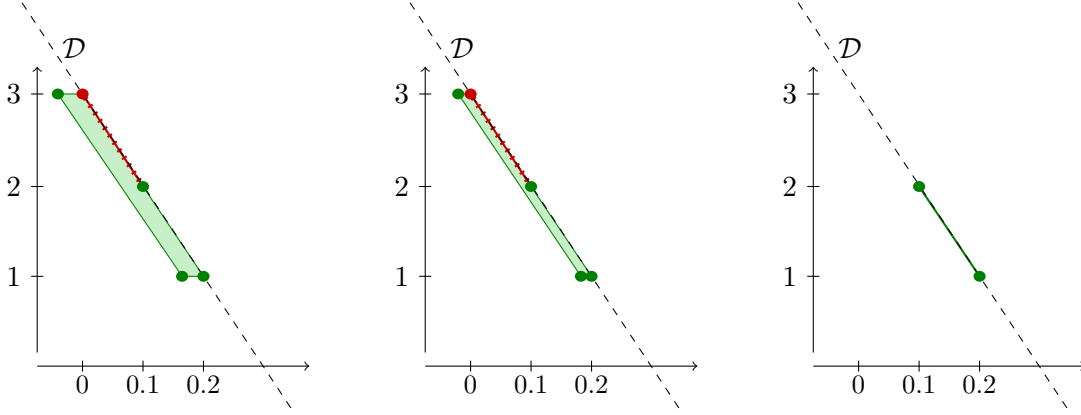


Figure 2: Representations of $\mathcal{M}_Q^\eta$ for $\eta = 0.1$ (left) and $\eta = 0.05$ (center) and of the limit $\mathcal{C}'_Q$ as $\eta \downarrow 0$ (right). The interiors are shaded in green. The endpoints of the two intervals and the point $(0.1, 2)^\top$ generating $\mathcal{M}_Q^\eta$ according to (45) are marked in green (for four of them, included) or in red (for the point not included). As a consequence, the semi-open red hatched interval is not part of $\mathcal{M}_Q^\eta$. The dashed line $\mathcal{D}$ is introduced in Step 1 of the proof.

Theorem 47 *In the setting above, where $\mathbf{p}^*$ is piecewise constant, there exists a sequence $(z_t)_{t \geq 1}$ in $\Delta(\mathcal{B})$ that satisfies*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(z_t, Q) = 0 \quad \text{but} \quad \liminf_{T \rightarrow \infty} d\left(\frac{1}{T} \sum_{t=1}^T \mathbf{m}^*(z_t), \mathcal{C}'_Q\right) > 0,$$

where $\mathcal{C}'_Q = \bigcap_{\eta > 0} \mathcal{M}_Q^\eta$.

Proof We decompose the proof of the counter-example into two steps. First, we provide closed-form expressions for the sets $\mathcal{M}_Q^\eta$ and for their limit $\mathcal{C}'_Q$, as illustrated in Figure 2. Second, we construct a sequence $(z_t)_{t \geq 1}$ that satisfies the claim.

Step 1: Structure. We show that for $0 < \eta < 0.2$,

$$\mathcal{M}_Q^\eta = \text{conv} \left(\left[\begin{pmatrix} -\sqrt{2/3}\eta \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right) \cup \left\{ \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \right\} \cup \left[\begin{pmatrix} 0.2 - \eta/\sqrt{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right] \right). \quad (45)$$

On the left and middle pictures of Figure 2, we illustrate what $\mathcal{M}_Q^\eta$ is the convex hull (in shaded green) of: the upper semi-open interval (the left endpoint is included and is marked in green while the right endpoint is not included is marked in red), the middle point (marked in green), and the lower closed interval (with both endpoints marked in green).

Based on this expression, we then show that $\mathcal{C}'_Q$ is a (small) segment,

$$\mathcal{C}'_Q \stackrel{\text{def}}{=} \bigcap_{\eta > 0} \mathcal{M}_Q^\eta = \left[\begin{pmatrix} 0.1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right]. \quad (46)$$

Step 1A: Proof of (45). It suffices to show that

$$\{\mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in Q^\eta\} = \left[\begin{pmatrix} -\sqrt{2/3}\eta \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right) \cup \left\{ \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \right\} \cup \left[\begin{pmatrix} 0.2 - \eta/\sqrt{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right], \quad (47)$$

where $Q^\eta = \{\mathbf{q} \in \Delta(\mathcal{B}) : d(\mathbf{q}, Q) \leq \eta\}$ denotes the closed η -neighborhood of Q . Recall that distances are measured in Euclidean norm.

We start with proving the reverse inclusion $\supseteq$ in (47). Since the distributions $(0, 0.5, 0.5)^\top$ and $(0, 0.2, 0.8)^\top$ belong to Q , we have that for all $0 < \eta' \leq \eta < 0.2$,

$$\mathbf{y}_{\eta'} = \begin{pmatrix} \sqrt{2/3}\eta' \\ 0.5 - \sqrt{2/3}\eta'/2 \\ 0.5 - \sqrt{2/3}\eta'/2 \end{pmatrix} \quad \text{and} \quad \mathbf{z}_{\eta'} = \begin{pmatrix} 0 \\ 0.2 - \eta'/\sqrt{2} \\ 0.8 + \eta'/\sqrt{2} \end{pmatrix} \quad \text{belong to} \quad Q^\eta. \quad (48)$$

The first distributions satisfy the conditions “ $q_1 > 0$ and $q_1 + q_2 > 0.2$ ”, thus $\mathbf{p}^*(\mathbf{y}_{\eta'}) = \boldsymbol{\delta}_2$ and

$$\{\mathbf{m}^*(\mathbf{y}_{\eta'}) : 0 < \eta' \leq \eta\} = \left[\begin{pmatrix} -\sqrt{2/3}\eta \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right) \subseteq \{\mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in Q^\eta\}.$$

Similarly, the second distributions satisfy the condition “ $q_1 + q_2 \leq 0.2$ ”, for the indicated η' but also for $\eta' = 0$, thus $\mathbf{p}^*(\mathbf{z}_{\eta'}) = \boldsymbol{\delta}_3$ and

$$\{\mathbf{m}^*(\mathbf{z}_{\eta'}) : 0 \leq \eta' \leq \eta\} = \left[\begin{pmatrix} 0.2 - \eta/\sqrt{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right] \subseteq \{\mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in Q^\eta\}.$$

Finally, the distribution $(0, 0.5, 0.5)^\top$ belongs to Q thus to Q^η , so that

$$\left\{ \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \right\} \subset \{\mathbf{m}^*(\mathbf{q}) : \mathbf{q} \in Q^\eta\}.$$

For the inclusion $\subseteq$ in (47), we first use the geometry of Q and of the simplex $\Delta(\mathcal{B})$ to show that

$$\forall \mathbf{q} \in Q^\eta, \quad q_1 \leq \sqrt{2/3}\eta \quad \text{and} \quad q_1 + q_2 \geq 0.2 - \eta/\sqrt{2}. \quad (49)$$

To prove these bounds, we fix $\mathbf{q} \in Q^\eta$ and denote by $\mathbf{q}'$ its projection onto Q in Euclidean norm; in particular, $\|\mathbf{q} - \mathbf{q}'\| = d(\mathbf{q}, Q)$.

For the upper bound on q_1 , we use that $q_1 + q_2 + q_3 = 1 = q'_2 + q'_3$ (since $q'_1 = 0$ by definition of Q), which yields in turn, together with $u^2 + v^2 \geq 2uv$,

$$q_1^2 = (q_2 - q'_2)^2 + (q_3 - q'_3)^2 + 2(q_2 - q'_2)(q_3 - q'_3) \geq 4(q_2 - q'_2)(q_3 - q'_3).$$

Therefore, by substituting first the inequality above, and then the equality above,

$$\begin{aligned} \sqrt{\frac{3}{2}} q_1 &= \sqrt{2q_1^2 - \frac{1}{2}q_1^2} \leq \sqrt{2q_1^2 - 2(q_2 - q'_2)(q_3 - q'_3)} \\ &= \sqrt{q_1^2 + (q_2 - q'_2)^2 + (q_3 - q'_3)^2} = \|\mathbf{q} - \mathbf{q}'\| = d(\mathbf{q}, Q) \leq \eta. \end{aligned}$$

For the lower bound on $q_1 + q_2$, we only need to prove it in the case where $q_1 + q_2 \leq 0.2$, which we assume next. The bound follows again from the equality $q'_3 - q_3 = q_1 + q_2 - q'_2$ and from the fact that $0 \leq q_1 + q_2 \leq 0.2 \leq q'_2$ (since $\mathbf{q}' \in Q$ and by definition of Q): therefore, $0 \leq 0.2 - (q_1 + q_2) \leq q'_2 - (q_1 + q_2)$, thus

$$\sqrt{2}(0.2 - (q_1 + q_2)) = \sqrt{2(0.2 - (q_1 + q_2))^2} \leq \sqrt{2(q'_2 - (q_1 + q_2))^2} \leq \underbrace{d(\mathbf{q}, Q)}_{\text{see below}} \leq \eta,$$

where the inequality marked above comes from

$$2(q_1 + (q_2 - q'_2))^2 = q_1^2 + (q_2 - q'_2)^2 + \underbrace{2q_1(q_2 - q'_2)}_{\leq 0} + \underbrace{(q_1 + q_2 - q'_2)^2}_{=(q_3 - q'_3)^2} \leq \|\mathbf{q} - \mathbf{q}'\|^2.$$

Finally, we resort to the bounds obtained in (49) together with the definition of $\mathbf{m}^*$, and get, for all $\mathbf{q} \in Q^\eta$,

$$\mathbf{m}^*(\mathbf{q}) = \begin{cases} \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} & \text{if } q_1 = 0 \text{ and } q_1 + q_2 > 0.2, \\ \begin{pmatrix} -q_1 \\ 3 \end{pmatrix} \in \left[\begin{pmatrix} -\sqrt{2/3}\eta \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right] & \text{if } q_1 > 0 \text{ and } q_1 + q_2 > 0.2, \\ \begin{pmatrix} q_1 + q_2 \\ 1 \end{pmatrix} \in \left[\begin{pmatrix} 0.2 - \eta/\sqrt{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right] & \text{if } q_1 + q_2 \leq 0.2, \end{cases}$$

which concludes the proof of the inclusion $\subseteq$ in (47).

Step 1B: Proof of (46). We consider the line $\mathcal{D}$ generated by the segment indicated in (46) and illustrated by the dashed line in Figure 2:

$$\mathcal{D} = \{\mathbf{c} : \phi(\mathbf{c}) = 0\}, \quad \text{where} \quad \phi : (x, y) \mapsto y - 3 + 10x.$$

We first note that the closed-form expression (45) of the $\mathcal{M}_Q^\eta$ shows that $\mathcal{C}'_Q \subseteq \mathcal{D}$. We now show that no point of the red hatched semi-open interval of Figure 2 (which is a subset of $\mathcal{D}$) belongs to any of the sets $\mathcal{M}_Q^\eta$, where $0 < \eta < 0.2$. Indeed, as illustrated in Figure 2, $\mathcal{M}_Q^\eta$ is included in the left hyperplane formed by $\mathcal{D}$, i.e., for all $\mathbf{c} \in \mathcal{M}_Q^\eta$, we have $\phi(\mathbf{c}) \leq 0$. Resorting to the Carathéodory's theorem and due to (45), any point in $\mathcal{M}_Q^\eta$ can be written as a convex combination of at most three points in

$$\mathcal{G}_Q^\eta = \left[\begin{pmatrix} -\sqrt{2/3}\eta \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right] \cup \left\{ \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \right\} \cup \left[\begin{pmatrix} 0.2 - \eta/\sqrt{2} \\ 1 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right].$$

Note that

$$\mathcal{G}_Q^\eta \cap \mathcal{D} = \mathcal{G}_Q^\eta \cap \{\mathbf{c} : \phi(\mathbf{c}) = 0\} = \left\{ \begin{pmatrix} 0.1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right\},$$

while $\phi(\mathbf{c}) < 0$ for all other points in $\mathcal{G}_Q^\eta$. Therefore, by the linearity of ϕ , only points in $\mathcal{M}_Q^\eta$ that can be written as convex combinations of $\mathcal{G}_Q^\eta \cap \mathcal{D}$ belong to $\mathcal{M}_Q^\eta \cap \mathcal{D}$:

$$\mathcal{M}_Q^\eta \cap \mathcal{D} = \text{conv}(\mathcal{G}_Q^\eta \cap \mathcal{D}) = \left[\begin{pmatrix} 0.1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right].$$

Thus, using the inclusion $\mathcal{C}'_Q \subseteq \mathcal{D}$ stated above,

$$\mathcal{C}'_Q = \mathcal{C}'_Q \cap \mathcal{D} = \bigcap_{\eta > 0} (\mathcal{M}_Q^\eta \cap \mathcal{D}) = \left[\begin{pmatrix} 0.1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0.2 \\ 1 \end{pmatrix} \right],$$

as announced.

Step 2: Construction of the sequence. We build a sequence $(\mathbf{z}_t)_{t \geq 1}$ that satisfies the claim of the counter-example, i.e., such that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) = 0 \quad \text{but} \quad \liminf_{T \rightarrow \infty} d\left(\frac{1}{T} \sum_{t=1}^T \mathbf{m}^*(\mathbf{z}_t), \mathcal{C}'_Q\right) > 0.$$

Following Step 1, we define the sequence in such a way that $\mathbf{z}_t$ is in a η_t -neighborhood of Q for $\eta_t = \sqrt{2}/(t+2)^2$, which is smaller than 0.2 for all $t \geq 1$. To do so, we consider a fixed distribution in Q , namely, $(0, 0.2, 0.8)^\top$, and introduce a vanishing perturbation of order $1/t^2$ to this distribution. Formally, we define the sequence $(\mathbf{z}_t)_{t \geq 1}$ as

$$\mathbf{z}_t = (1/(t+2)^2) \boldsymbol{\delta}_1 + 0.2 \boldsymbol{\delta}_2 + (0.8 - 1/(t+2)^2) \boldsymbol{\delta}_3.$$

By construction, the sequence $(\mathbf{z}_t)_{t \geq 1}$ satisfies $d(\mathbf{z}_t, Q) \leq \sqrt{2}/(t+2)^2$ for all $t \geq 1$, so that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T d(\mathbf{z}_t, Q) = 0.$$

We now use the continuity of the point-to-set distance and show that

$$\lim_{T \rightarrow \infty} d\left(\frac{1}{T} \sum_{t=1}^T \mathbf{m}^*(\mathbf{z}_t), \mathcal{C}'_Q\right) = \left\| \begin{pmatrix} 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 0.1 \\ 2 \end{pmatrix} \right\| = \sqrt{1.01} > 0.$$

Indeed, each distribution $\mathbf{z}_t = (z_{t,1}, z_{t,2}, z_{t,3})^\top$ is such that $z_{t,1} > 0$ and $z_{t,1} + z_{t,2} > 0.2$; thus, for all $t \geq 1$, the definitions of $\mathbf{p}^*$ and $\mathbf{m}^*$ imply $\mathbf{p}^*(\mathbf{z}_t) = \boldsymbol{\delta}_2$ and

$$\mathbf{m}^*(\mathbf{z}_t) = \begin{pmatrix} -1/(t+2)^2 \\ 3 \end{pmatrix}, \quad \text{so that} \quad \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{m}^*(\mathbf{z}_t) = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

The average reward converges to the point $(0, 3)^\top$, and the closest point in $\mathcal{C}'_Q$ is $(0.1, 2)^\top$, as follows from the closed-form expression of $\mathcal{C}'_Q$ derived in Step 1 (see also Figure 2). This concludes the proof. $\blacksquare$

Remark 48 *The sequence $(\mathbf{z}_t)_{t \geq 1}$ in Bernstein et al. (2014, Corollary 2) is generated by a calibrated forecaster, typically involving randomized predictions (Oakes, 1985; Dawid, 1985). In contrast and with no loss of generality, Step 2 of the proof of the counter-example uses a deterministic sequence.*

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